A Hierarchy of Risk Certificates for Forecasting Representations

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Abstract

We study whether a fixed forecasting representation can meet an expected squared-error tolerance under a decoder sensitivity budget. For bounded scalar targets, we construct normalized minimum-risk scores on blocks of observations. The size-two score is a pairwise obstruction; expected scores increase with block size and converge to optimal Lipschitz risk under unconditional iid sampling and compact metric representation support. Exact rare-conflict and noisy-support laws show that no fixed block size is uniformly sufficient. Independent-block calibration and estimated-mass stratification turn these population bounds into finite-sample lower certificates. Combined with upper bounds for fitted forecasts, they can certify minimum feasible cost among declared representation classes. Learned Kernel Affine Hull Machine (KAHM) dictionaries provide an expandable implementation. Fresh calibration of forty frozen banks shows larger blocks excluding classes when pairs are inconclusive, while stratification and full-sample empirical minimization can give stronger decisions. Broader forecasting studies retain task-dependent accuracy and unresolved cases. The certificates assess a representation, metric, sensitivity budget, target, and evaluation law jointly; they do not identify every positive lower bound as removable information loss. Informative natural-data class exclusion remains unestablished.

Keywords: Kernel Affine Hull Machines, time-series forecasting, Lipschitz regression, risk certification, representation selection

1 Introduction

Forecasting representations are commonly assessed through the errors of fitted decoders. Such comparisons do not determine whether another decoder could make a simpler representation meet the same accuracy requirement. We study how to certify that distinction. An upper risk bound establishes the accuracy of a fitted forecast; a lower risk bound can exclude every decoder in a declared representation class. Together they can establish when a more costly representation is necessary. All comparisons use the same observed target and evaluation law, with decoder sensitivity budgets declared explicitly. The added conclusion is necessity within a declared family: the least costly accepted fit is certified to have minimum feasible cost when every cheaper class is excluded. This conclusion can hold even when upper-only selection returns the same fitted forecast.

Kernel Affine Hull Machines (KAHMs) provide our main worked example. Their established local reconstruction models map histories to soft associations with reference groups (Kumar et al., 2024, 2025; Kumar and Kargaran, 2026). We retain these models and append feature blocks under a sum-of-block-norms decoder budget. This preserves earlier feasible forecasts and makes exact block duplication immaterial to the class. The problem is to assess whether these learned

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coordinates support a forecasting tolerance, and whether an added block is necessary to attain it. The certificates apply to other fixed representations as well; no theorem assumes that reconstruction affinity establishes predictive sufficiency or exact linear feature evolution.

A basic obstruction arises when two targets are farther apart than the sensitivity budget permits their predictions to be. Related pairwise Lipschitz regression arguments are established (Akash et al., 2021). Averaging the squared positive part of this discrepancy over conditionally independent pairs yields a risk lower bound. This inexpensive statistic can be exact, but it can also miss almost all of the attainable class risk. We ask how to reduce that population slack while accounting separately for uncertainty from a finite calibration sample.

Our main construction replaces pairs by blocks of observations. Within each block, we minimize squared error over predictions satisfying all pairwise Lipschitz constraints and normalize by one less than the block size. The size-two statistic is exactly the pairwise obstruction. For scalar targets, the expectations of these statistics form a nondecreasing hierarchy of lower bounds. Under unconditional iid sampling, bounded targets, and compact representation support, they converge to the optimal Lipschitz risk. Larger blocks capture compatibility constraints that individual pairs cannot express. They also require more optimization and leave fewer independent block scores for calibration. Population improvement therefore does not imply a uniformly stronger finite-sample certificate. Monotonic lower expectations of ordinary sample-average minima are established (Mak et al., 1999). Here the centered normalization connects the Lipschitz problem exactly to the pair score, and the rare-conflict and noisy-support laws quantify the information missed by finite blocks.

Conditional pairing offers a complementary approach. Comparisons within a predefined history stratum can expose conflicts that global pairing rarely encounters. Reweighting by stratum probabilities preserves the marginal risk; calibration accounts for random counts and estimated masses. A lower bound nevertheless assesses the entire representation–decoder class: positive risk may reflect information loss, irreducible future noise, or the sensitivity constraint. We do not identify these causes from a lower endpoint alone.

The paper makes three contributions.

1. **A centered hierarchy and its population limits.** We connect the pair witness to normalized minimum-risk problems on larger blocks, establish monotonicity and consistency under the stated conditions, and identify exact and arbitrarily loose pairwise cases.

2. **Representation decisions from lower and upper bounds.** Independent block scores and stratification give class-wide lower certificates. Combined with fitted-risk upper bounds and retained-block geometry, they support accuracy, improvement, and minimum feasible cost statements for a frozen family. The concentration inequalities and simultaneous-selection principles are established components.

3. **Tests of usefulness and failure.** Exact-risk comparisons separate population slack, precision, and optimization cost. Fresh calibration on 40 frozen KAHM feature banks shows larger blocks changing class-exclusion decisions, while retaining cases where full-sample empirical minimization is stronger. Controlled refinement and broader synthetic, pen-trajectory, and speech studies assess accuracy, scaling, and calibration limits.

The evidence separates informative decisions from unresolved ones. Empirical-minimum calibration is often tighter than pairwise calibration, and KAHM accuracy depends on the task and decoder. Useful class exclusion on a constructed mechanism does not establish its prevalence on

Question	Kumar and Kargaran (2026)	Present work
Certificate target	Successor soft coordinates	Fixed observed future outputs
Decoder class	Spectral-norm-bounded linear coordinate predictors	General Lipschitz oracle; retained-block affine fits
Lower information	Class-centered variation; empirical-minimum comparator	Positive-part pair slack; normalized monotone block hierarchy
Representation decisions	Accuracy exclusion for a fixed coordinate map and norm limit	Simultaneous class exclusion and fitted-model acceptance; minimum feasible cost among declared alternatives

Table 1: Scope relative to arXiv:2609.32652v3. KAHM reconstruction, independent evaluation, and standard concentration are shared foundations. The scalar block consistency result requires the conditions in Theorem 4.

natural data. All confidence statements concern expected squared error under the declared sampling law. Sections 3–4 develop the population bounds; Sections 5–6 connect them to fixed forecasting procedures and calibrated decisions.

2 Related work

Geometric representations and Koopman prediction. KAHMs learn reconstruction models in reproducing kernel Hilbert spaces (RKHSs) (Kumar et al., 2024, 2025). We use these established reconstruction models to construct forecasting features. Soft state aggregation (Singh et al., 1994), approximate state abstraction (Abel et al., 2016), and task-oriented latent transition models (Gelada et al., 2019) likewise distinguish a representation from its predictive use. Extended dynamic mode decomposition approximates Koopman operators on finite dictionaries (Williams et al., 2015); learned dictionaries (Li et al., 2017) and RKHS regression bounds (Kostic et al., 2022) provide related forecasting frameworks. Basis-invariant consistency diagnostics (Haseli and Cortés, 2023), multistep projection-error guarantees and refinement (Conradie et al., 2026), and residual-based dictionary learning (Coote and Colbrook, 2026) address dictionary quality. Our decisions instead compare expected observed-output error with a declared tolerance while preserving a common target and sampling law.

Closest KAHM prediction-limit analysis. The earlier analysis of Kumar and Kargaran (2026), in its cited third version, provides class-centered lower certificates, empirical-minimum comparisons, fitted-risk upper bounds, and independent evaluation. It also gives an exact four-state example and independent dynamical confirmation. We reuse its KAHM construction and sampling principle; neither certificates for KAHM nor independent confirmation is new here. Table 1 states the substantive differences in the certificates. That work also includes common-target forecasting comparisons; the present extension concerns observed-output certificates and decisions over a retained family. Their reconstruction-score margin bounds concern agreement with reference labels, which does not imply predictive sufficiency.

Pairwise bounds and Lipschitz regression. The paired-residual argument underlying Akash et al. (2021, Lemma 2) uses target separation and Lipschitz control to lower-bound an adversary's mean

squared error (MSE). We use a conditional squared-positive-part score and examine its exactness, population slack, and calibration for class-level forecasting decisions under an unchanged marginal risk. Finite-support Lipschitz regression can be posed as a constrained optimization problem (Gottlieb et al., 2017); we use this class directly for an empirical-minimum comparator. Model-class falsification (Müller et al., 2025) and minimal-empirical-risk bounds (Wu and Yang, 2026) provide broader context. The Gamma test uses local pair differences to estimate regression noise (Evans and Jones, 2002). Lipschitz set-membership identification (Milanese and Novara, 2004), trajectory invalidation (Jin et al., 2020), and corruption-robust prediction ranges (Blum and Nguyen, 2026) concern related feasibility questions with different sampling and error criteria.

Empirical minima and optimization over blocks. Monotonic lower expectations of sample-average optimal values are established in stochastic programming (Mak et al., 1999). Repeated-batch estimation (Linderoth et al., 2006) and non-asymptotic optimal-value confidence bounds (Guigues et al., 2017) provide further precedents. Optimization of empirical U-statistics and its computational–statistical tradeoffs are also established (Clémençon et al., 2016). Our construction applies these principles to scalar Lipschitz prediction. Translation of the fitted values justifies a residual-variance normalization, rather than a generic correction of empirical risk. It produces the exact positive-part pair score at size two, a monotone hierarchy converging under compactness, and exact rare-conflict and occupancy laws describing finite-block slack. For each fixed decoder the residual-difference objective is a U-statistic; its optimized value is not generally a U-statistic with a fixed kernel. Calibration instead uses standard bounded-mean inequalities on independent blocks.

Calibration, refinement, and dependence. We use classical bounded-mean inequalities (Hoeffding, 1963) and empirical Bernstein and standard-deviation bounds (Maurer and Pontil, 2009). The simultaneous testing principle follows Learn then Test (Angelopoulos et al., 2025). Our group norm is established in group selection and multiple-kernel learning (Bach, 2008); here its max-block dual metric gives a comparable sensitivity budget as dictionaries grow. Soft gating and residual fitting also have established precedents (Jordan and Jacobs, 1994; Bühlmann and Yu, 2003). The additional cost conclusion comes from jointly excluding cheaper classes and accepting a fitted model.

Within-episode dependence is allowed because concentration is applied to independent episode summaries. A chronological split of one dependent record does not provide that independence. Forecasting bounds under other sampling assumptions (McDonald et al., 2017) and dependence-aware conformal analysis (Oliveira et al., 2024) illustrate the additional conditions needed; their guarantees are not consequences of the present theorem.

3 Pairwise obstructions and their limitations

This section defines the representation risk and derives its pairwise obstruction. All comparisons use the same distribution of histories and observed forecasting targets. The histories need not form a Markov state, and neither stationarity nor exact linear feature evolution is assumed.

3.1 A directly estimable pairwise obstruction

Let $X \in \mathcal{X}$ be a random available history, let $Y \in \mathbb{R}^q$ be its future target, and let $U = \Gamma(X) \in \mathcal{U}$ be a fixed, measurable representation. Here $\mathcal{X}$ and $\mathcal{U}$ denote the history and representation spaces. A measurable pseudometric $d : \mathcal{U} \times \mathcal{U} \to [0, \infty)$ is symmetric, vanishes on the diagonal, and satisfies the triangle inequality; unlike a metric, it may assign zero distance to distinct points. For a sensitivity

budget $\kappa \geq 0$, define

$$\mathcal{G}_\kappa(d) = \{g : \mathcal{U} \to \mathbb{R}^q \text{ measurable} : \|g(u) - g(v)\|_2 \leq \kappa d(u,v) \text{ for all } u, v \in \mathcal{U}\},$$
$$R^*_\kappa(\Gamma, d) = \inf_{g \in \mathcal{G}_\kappa(d)} \mathbb{E}\|Y - g(\Gamma(X))\|_2^2.$$

Thus κ limits how much a forecast can change when its representation changes, and R^*_κ is the best population squared-error risk permitted by that limit. For any predictor $f : \mathcal{X} \to \mathbb{R}^q$, write $R(f) = \mathbb{E}\|Y - f(X)\|_2^2$. The representation, pseudometric, and budget are fixed before the calibration data are observed. The class $\mathcal{G}_\kappa(d)$ may contain more decoders than the fitted family; a lower bound on its optimal risk therefore also applies to that family. When the representation and pseudometric are clear, we abbreviate $R^*_\kappa(\Gamma, d)$ to R^*_κ and similarly suppress the arguments of the witness defined below.

What the oracle risk measures. For $\mathbb{E}\|Y\|_2^2 < \infty$, write $M_X = \mathbb{E}[Y \mid X]$ and $M_U = \mathbb{E}[Y \mid U]$. The standard orthogonality of conditional expectation gives

$$R^*_\kappa(\Gamma, d) = \underbrace{\mathbb{E}\|Y - M_X\|_2^2}_{\text{irreducible history risk}} + \underbrace{\mathbb{E}\|M_X - M_U\|_2^2}_{\text{representation information loss}} + \underbrace{\inf_{g \in \mathcal{G}_\kappa(d)} \mathbb{E}\|M_U - g(U)\|_2^2}_{\text{sensitivity approximation}}. \tag{1}$$

Indeed, U is measurable with respect to X, so $\mathbb{E}[M_X \mid U] = M_U$. For every finite-risk g, the three terms in $Y - g(U) = (Y - M_X) + (M_X - M_U) + (M_U - g(U))$ are pairwise orthogonal in L^2. Expanding and taking the infimum proves the identity; constant decoders ensure that this infimum is finite. This standard decomposition is not a new result. It shows why a positive risk lower bound alone cannot distinguish information loss from future noise or a restrictive budget. Its information-loss term concerns the conditional mean under squared error, not full conditional-distribution or Markov sufficiency. Restricting the decoder further to a restricted affine family can add another approximation gap, as the example at the end of this section demonstrates.

Any training-fitted preprocessing is part of Γ. The population results below do not require the input coordinates to lie in $[-1, 1]$. For the finite-sample certificates, the deterministic range assumption concerns the target and prediction loss. A common value of κ across preprocessing choices limits sensitivity in each resulting feature metric; it does not impose a common Lipschitz constant in the original input units. Each choice therefore specifies its own representation class and must be frozen before calibration.

To define the pairwise statistic, first draw a conditioning variable A. Then draw (X, Y) and (X', Y') independently from the same conditional law given A, and set $U = \Gamma(X)$ and $U' = \Gamma(X')$. Each copy must have the original marginal law of (X, Y). Taking A to be constant gives an ordinary independent pair. With $z_+ = \max\{z, 0\}$, define

$$W_{\kappa,A}(\Gamma, d) = \frac{1}{2}\mathbb{E}\left[\left(\|Y - Y'\|_2 - \kappa d(U, U')\right)_+^2\right]. \tag{2}$$

The expectation includes both the draw of A and the two conditional draws. We call $W_{\kappa,A}$ a *pair witness*, or *obstruction*: it measures target separation that a decoder with budget κ cannot reproduce from the representation separation. The variable A specifies a sampling construction. Arbitrary nearest-neighbor matching need not have this conditional independence property.

Theorem 1 (Pairwise prediction obstruction) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$. Then*

$$R_\kappa^*(\Gamma, d) \geq W_{\kappa,A}(\Gamma, d). \tag{3}$$

For a representation space $\mathcal{U} \subseteq \mathbb{R}^p$ with $d(u,v) = \|u - v\|_2$, where p is its ambient dimension, assume also $\mathbb{E}\|U\|_2^2 < \infty$. Define

$$\sigma_{Y|A}^2 = \mathbb{E}\|Y - \mathbb{E}[Y \mid A]\|_2^2, \qquad \tau_{U|A}^2 = \mathbb{E}\|U - \mathbb{E}[U \mid A]\|_2^2,$$

where $\sigma_{Y|A}$ and $\tau_{U|A}$ are the nonnegative square roots. These are scalar, averaged conditional variances, rather than functions of a particular value of A. Then

$$W_{\kappa,A}(\Gamma, \|\cdot\|_2) \geq (\sigma_{Y|A} - \kappa\tau_{U|A})_+^2. \tag{4}$$

Both inequalities use the same conditioning variable and evaluation law.

Proof Fix $g \in \mathcal{G}_\kappa(d)$ with finite risk and define the two residuals $r = Y - g(U)$ and $r' = Y' - g(U')$. Since $Y - Y' = r - r' + g(U) - g(U')$, the triangle inequality gives

$$\|Y - Y'\|_2 \leq \|r - r'\|_2 + \kappa d(U, U'), \qquad (\|Y - Y'\|_2 - \kappa d(U, U'))_+ \leq \|r - r'\|_2.$$

Conditional on A, the residuals are independent and identically distributed. Expanding the squared norm therefore yields

$$\begin{aligned}\mathbb{E}[\|r - r'\|_2^2 \mid A] &= 2\mathbb{E}[\|r\|_2^2 \mid A] - 2\mathbb{E}[r \mid A]^\top \mathbb{E}[r' \mid A] \\ &= 2\left\{\mathbb{E}[\|r\|_2^2 \mid A] - \|\mathbb{E}[r \mid A]\|_2^2\right\}.\end{aligned}$$

Square the preceding pointwise bound, take expectations, and divide by two. This gives $W_{\kappa,A} \leq \mathbb{E}\|r\|_2^2 = R(g \circ \Gamma)$. Taking the infimum over g proves (3); predictors with infinite risk satisfy the bound trivially.

For (4), set $a = \|Y - Y'\|_2$ and $b = \|U - U'\|_2$. Pointwise, $a \leq (a - \kappa b)_+ + \kappa b$. For a scalar random variable V, the notation $\|V\|_{L^2} = (\mathbb{E}V^2)^{1/2}$ denotes its root second moment under the pairing law. The triangle inequality in L^2 gives

$$\|(a - \kappa b)_+\|_{L^2} \geq (\|a\|_{L^2} - \kappa\|b\|_{L^2})_+.$$

The same conditional-independence calculation gives $\mathbb{E}a^2 = 2\sigma_{Y|A}^2$ and $\mathbb{E}b^2 = 2\tau_{U|A}^2$. Substitute these identities, square, and divide by two to obtain (4). ■

Proposition 2 (An exact decomposition of witness slack) *For any admissible decoder g with finite risk, put $r = Y - g(U)$, $r' = Y' - g(U')$, and $v = (\|Y - Y'\|_2 - \kappa d(U, U'))_+$. Then*

$$R(g \circ \Gamma) - W_{\kappa,A} = \mathbb{E}\|\mathbb{E}[r \mid A]\|_2^2 + \frac{1}{2}\mathbb{E}\big[\|r - r'\|_2^2 - v^2\big]. \tag{5}$$

Both terms are nonnegative. Consequently, g attains the witness, meaning $R(g \circ \Gamma) = W_{\kappa,A}$, if and only if $\mathbb{E}[r \mid A] = 0$ almost surely and $\|r - r'\|_2 = v$ almost surely. Whenever $r \neq r'$, the latter condition requires $\|g(U) - g(U')\|_2 = \kappa d(U, U')$. It also requires the decoder difference to be a nonnegative scalar multiple of the residual difference; the scalar may be zero.

Proof Conditional independence gives $\frac{1}{2}\mathbb{E}\|r-r'\|_2^2 = R(g\circ\Gamma)-\mathbb{E}\|\mathbb{E}[r \mid A]\|_2^2$. Subtract the witness and use $v \le \|r-r'\|_2$ from the preceding proof. Equality requires both nonnegative terms to vanish. If $r \neq r'$, it also requires equality in $\|Y-Y'\|_2 \le \|r-r'\|_2+\|g(U)-g(U')\|_2 \le \|r-r'\|_2+\kappa d(U,U')$, giving the stated conditions. ■

The witness directly estimates a squared distance discrepancy. It does not estimate two variances separately and then subtract their square roots. This distinction can matter when little input variation remains in the representation. The certificates below use concentration for bounded means; they establish no new minimax estimation rate. Taking a square root of a risk bound can still produce slow convergence near zero. Moreover, (4) compares certificates under a common pairing law. It does not compare pairing at a common time origin with conditioning on a different grouping, such as a reference-class label $c(X)$, where c assigns a history to a group.

Proposition 3 (Exactness on a two-atom law) *Suppose (U,Y) takes only two values: (u_0,y_0) with probability $p \in (0,1)$ and (u_1,y_1) with probability $1-p$. Use Euclidean representation distance and an unconditioned independent pair, and omit the constant conditioning variable from the notation. Then*

$$R^*_\kappa = W_\kappa = p(1-p)\big(\|y_1-y_0\|_2 - \kappa\|u_1-u_0\|_2\big)_+^2. \tag{6}$$

An affine map attains this optimum. If forecasts must lie in a nonempty closed convex set containing y_0, y_1, projecting that map onto the set preserves both the optimum and the Lipschitz budget.

Proof Write $a = \|y_1-y_0\|_2$, $b = \|u_1-u_0\|_2$, and $\mu = py_0+(1-p)y_1$. The unequal ordered pairs have total probability $2p(1-p)$, whereas equal pairs contribute zero. Thus (2) gives the right side of (6) as a lower bound.

To attain the bound, first consider $a = 0$: the constant forecast y_0 has zero risk. If $a > 0$, let $v = (y_1-y_0)/a$ be the unit target direction and let $s = \min\{a, \kappa b\}$ be the allowed fitted separation. Choose

$$f_0 = \mu - (1-p)sv, \qquad f_1 = \mu + psv.$$

Then $y_0 - f_0 = -(1-p)(a-s)v$ and $y_1 - f_1 = p(a-s)v$, so their weighted squared error is $p(1-p)(a-s)^2$, exactly the proposed bound. If $b = 0$, then $s = 0$ and the constant map μ realizes both fitted values. If $b > 0$, use

$$g(u) = f_0 + \frac{\langle u-u_0, u_1-u_0\rangle}{b^2}(f_1-f_0).$$

Its linear part has operator norm $\|f_1-f_0\|_2/b = s/b \le \kappa$, so it is an admissible affine decoder. Both fitted values lie on the segment joining y_0 and y_1. Projection onto the stated convex set fixes them and is nonexpansive, meaning that it cannot increase distances. ■

Exactness also holds on continuous supports. Let T be a square-integrable real random variable, set $U = T$ and $Y = b + \beta Tv$, and assume $b, v \in \mathbb{R}^q$, $\|v\|_2 = 1$, and $\beta \ge 0$. Here β is the target's slope, and $\mathrm{Var}(T) = \mathbb{E}(T-\mathbb{E}T)^2$. For independent T, T' and $d(t,t') = |t-t'|$, the pair discrepancy is $(\beta-\kappa)_+|T-T'|$. Since $\mathbb{E}(T-T')^2 = 2\,\mathrm{Var}(T)$, its witness is $(\beta-\kappa)_+^2\,\mathrm{Var}(T)$. The affine decoder

$$g(t) = b + \beta\mathbb{E}[T]v + \min\{\beta,\kappa\}(t-\mathbb{E}[T])v$$

has Lipschitz constant at most κ and residual $(\beta-\kappa)_+(T-\mathbb{E}T)v$. Its risk equals the witness, so $W_\kappa = R^*_\kappa$ also in this continuous-support example.

Two sources of population looseness. Even with independent unconditioned pairs, the witness need not approximate the optimum by a fixed relative factor. Let $\kappa = 1$ and let (U, Y) take values $(3, 0), (0, -1), (1, 1)$ with probabilities $1 - 2\varepsilon, \varepsilon, \varepsilon$, where $0 < \varepsilon < 1/2$. Only the two rare atoms have a positive discrepancy, so $W_\kappa = \varepsilon^2$. The fitted values $g(0) = -1/2$, $g(1) = 1/2$, and $g(3) = 0$ attain risk $\varepsilon/2$. Linear interpolation between these points, with constant values outside $[0, 3]$, defines a 1-Lipschitz decoder. Conversely, any admissible fitted values $v_0 = g(0)$ and $v_1 = g(1)$ satisfy $|v_1 - v_0| \le 1$. The minimum of $(-1 - v_0)^2 + (1 - v_1)^2$ under this constraint is $1/2$, attained at $v_0 = -1/2$ and $v_1 = 1/2$. The two rare atoms alone therefore contribute at least $\varepsilon/2$ to every decoder's risk. Thus $R^*_\kappa = \varepsilon/2$ and $W_\kappa/R^*_\kappa = 2\varepsilon$ can approach zero. The same example lies in Δ_2 after mapping u to $(u/3, 1 - u/3)$ and setting $\kappa = 3/\sqrt{2}$. Stratification below replaces the quadratic probability of encountering this rare pair by the probability of its stratum.

A separate gap comes from comparing the fitted affine family with all Lipschitz decoders. For equally likely $X \in \{-1, 0, 1\}$, take $\Phi(x) = ((x + 1)/2, (1 - x)/2)^\top$, $Y = X^2$, and $\kappa = \sqrt{2}$, with Euclidean distance between feature vectors. The target satisfies the Lipschitz inequalities on this support and has a scalar Lipschitz extension, so $W_\kappa = R^*_\kappa = 0$. An affine function of $\Phi(x)$ is affine in x; clipping it to $[0, 1]$ preserves monotonicity. A nondecreasing forecast cannot fit the target pair $(1, 0)$ at $x = -1, 0$: its minimum sum of squared errors on that pair is $1/2$, attained when both fitted values are $1/2$. A nonincreasing forecast has the same obstruction on the target pair $(0, 1)$ at $x = 0, 1$. Averaging over the three atoms gives MSE at least $1/6$ for every clipped affine forecast. The unprojected affine optimum is the constant $2/3$, with MSE $2/9$. Thus a sharp lower bound for all Lipschitz decoders need not describe the optimum of a restricted affine family.

4 A hierarchy of consistent block witnesses

The pair witness can lose information because its optimal fitted values are chosen separately for each pair. Requiring one set of fitted values to respect every distance in a larger block removes some of this freedom. We now make this observation precise for a *scalar* target. The result connects the pair obstruction to the optimal Lipschitz risk, while exposing the statistical cost of using larger blocks.

Fix the representation, pseudometric, and sensitivity budget from Section 3. Let $(U_i, Y_i)_{i=1}^b$ be independent copies from the full evaluation law, where $b \ge 2$, and assume $Y \in [y_-, y_+]$ almost surely. Write $D = y_+ - y_-$. Define

$$T_b = \frac{1}{b-1} \min_{v \in \mathbb{R}^b} \sum_{i=1}^{b} (Y_i - v_i)^2, \qquad \mathcal{W}_b = \mathbb{E}T_b, \tag{7}$$
$$\text{subject to } |v_i - v_j| \le \kappa d(U_i, U_j) \quad (i, j \le b).$$

The feasible set contains every constant vector. The minimum therefore exists and is unique. Its value is unchanged if the fitted values are restricted to $[y_-, y_+]$, because clipping preserves feasibility and cannot increase squared error. The factor $1/(b-1)$ is essential: if a_b denotes the usual minimum empirical MSE, then $T_b = ba_b/(b-1)$. This correction is justified by the freedom to translate every fitted value by the same constant; it is not a generic correction for arbitrary empirical-risk minima.

Theorem 4 (Monotone block hierarchy) *Under these assumptions, the following statements hold.*

(i) The first member is exactly the pair score:

$$T_2 = \tfrac{1}{2}\big(|Y_1 - Y_2| - \kappa d(U_1, U_2)\big)_+^2.$$

Consequently $\mathcal{W}_2 = W_{\kappa,A}$ when A is constant.

(ii) For every $b \geq 2$,

$$0 \leq T_b \leq B_b^{\mathrm{blk}}, \qquad B_b^{\mathrm{blk}} = \frac{D^2\lfloor b^2/4 \rfloor}{b(b-1)}, \qquad \mathcal{W}_b \leq R_\kappa^*. \tag{8}$$

(iii) If $2 \leq m \leq b$ and $T_m(S)$ is the score computed from a subset $S \subseteq \{1, \ldots, b\}$ of size m, then

$$\binom{b}{m}^{-1} \sum_{|S|=m} T_m(S) \leq T_b. \tag{9}$$

In particular, $\mathcal{W}_2 \leq \mathcal{W}_3 \leq \cdots \leq R_\kappa^*$.

(iv) If the representation takes values in a compact metric space and d is that metric, then $\mathcal{W}_b \uparrow R_\kappa^*$ as $b \to \infty$.

Proof For two observations, let $\Delta = |Y_1 - Y_2|$ and $c = \kappa d(U_1, U_2)$. If $\Delta \leq c$, both targets can be fitted exactly. Otherwise the optimal fitted separation is c, with the two residuals having magnitude $(\Delta - c)/2$ and opposite signs. Their squared sum is $(\Delta - c)^2/2$, proving (i).

For any fixed admissible decoder g of finite risk, put $r_i = Y_i - g(U_i)$ and $\bar{r} = b^{-1}\sum_i r_i$. The translated values $v_i = g(U_i) + \bar{r}$ are feasible. Thus

$$T_b \leq \frac{1}{b-1}\sum_i (r_i - \bar{r})^2, \qquad \mathcal{W}_b \leq \mathrm{Var}(Y - g(U)) \leq R(g \circ \Gamma).$$

Taking the infimum over g proves the population bound. Using the constant fitted value $\bar{Y}$ instead bounds T_b by the unbiased sample variance of the targets. This variance is a convex function of each target coordinate, so its maximum over $[y_-, y_+]^b$ occurs at interval endpoints. If k coordinates equal y_+ and $b - k$ equal y_-, its value is $D^2 k(b-k)/\{b(b-1)\}$. Maximizing over integer k proves the range bound.

For (iii), let v minimize the full-block objective and set $r_i = Y_i - v_i$. Translation invariance implies $\sum_i r_i = 0$: otherwise translating v by the residual mean would strictly lower the objective. Hence

$$T_b = \frac{1}{b(b-1)}\sum_{i<j} (r_i - r_j)^2. \tag{10}$$

On a subset S, restricting v and translating by that subset's residual mean gives feasible fitted values. Therefore

$$T_m(S) \leq \frac{1}{m(m-1)} \sum_{i<j,\ i,j\in S} (r_i - r_j)^2.$$

A uniformly chosen m-subset contains a given pair with probability $m(m-1)/\{b(b-1)\}$. Averaging proves (9); taking expectations proves monotonicity. This is an average-over-subsets statement, not an ordering between arbitrary realized blocks of different sizes.

To prove (iv), any feasible vector extends to a scalar Lipschitz function by $g_v(u) = \min_i\{v_i + \kappa d(u, U_i)\}$. The triangle inequality and the finite constraints show that this function is κ-Lipschitz and equals v_i at U_i. Clipping it to $[y_-, y_+]$ preserves those properties. Thus a_b is the empirical minimum over the same bounded Lipschitz class as the population problem. On a compact metric space this class is compact in the uniform norm: it is uniformly bounded and equicontinuous. For every integer $m \geq 1$, choose a finite uniform $1/m$-net. The ordinary strong law holds simultaneously on the countable union of these nets, on an event of probability one. Since $|(y - f(u))^2 - (y - g(u))^2| \leq 2D\|f - g\|_\infty$ for interval-valued f, g, approximation by these nets gives uniform convergence over the entire class. Therefore $a_b \to R^*_\kappa$ almost surely. Multiplication by $b/(b-1)$ and the bound $T_b \leq D^2/2$ give $\mathcal{W}_b \to R^*_\kappa$ by dominated convergence. ■

The result strengthens the population witness without changing the decoder class. In particular, every block score dominates the average of its pair scores. The additional information is joint consistency of the fitted values across pairs, rather than a new concentration inequality. Optimization of empirical U-statistics is established (Clémençon et al., 2016); equation (10) gives the residual-difference form of this construction. Monotonic expected sample-average minima (Mak et al., 1999) and independent-batch lower estimation (Linderoth et al., 2006) are established precedents. The specific connection here is the centered normalization, its exact pair-obstruction member, and its finite-block limitations for Lipschitz prediction. The optimized score is calibrated as a bounded block statistic, without treating its data-dependent minimizer as a fixed U-statistic kernel.

4.1 Exact gaps and the limits of fixed block sizes

Consistency for a fixed distribution does not provide a universal choice of b. Two exact calculations distinguish rare predictive conflicts from the capacity to interpolate many distinct inputs.

Proposition 5 (Two exact block-size laws) *For the three-atom distribution from Section 3, with $(U, Y) = (3, 0), (0, -1), (1, 1)$ of probabilities $1 - 2\varepsilon, \varepsilon, \varepsilon$ and $\kappa = 1$,*

$$R^*_\kappa = \frac{\varepsilon}{2}, \qquad \mathcal{W}_b = \frac{2b\varepsilon - 1 + (1 - 2\varepsilon)^b}{4(b-1)}. \tag{11}$$

For every fixed b, $\mathcal{W}_b/R^*_\kappa \to 0$ as $\varepsilon \downarrow 0$.

Separately, let U be uniform on $M \geq 1$ points whose pairwise distances are at least D/κ, where $D, \kappa > 0$, and let $Y \in \{0, D\}$ be independent of U and equiprobable. Then

$$R^*_\kappa = \frac{D^2}{4}, \qquad \mathcal{W}_b = \frac{D^2}{4(b-1)}\left[b - M\{1 - (1 - 1/M)^b\}\right]. \tag{12}$$

For fixed b, $\mathcal{W}_b \to 0$ as $M \to \infty$, while the optimal risk stays $D^2/4$.

Proof In the first law, let N_- and N_+ count the two rare atoms in a block, and put $N = N_- + N_+$. The common atom can be fitted at zero without restricting the optimum at the rare atoms. Their constrained weighted least-squares error is N_-N_+/N when $N > 0$, and zero otherwise. Conditional

on N, N_- is binomial with parameters N and $1/2$, so the conditional mean of this error is $(N-1)/4$ for $N \geq 1$. Since $N \sim \mathrm{Binomial}(b, 2\varepsilon)$,

$$(b-1)\mathcal{W}_b = \tfrac{1}{4}\{\mathbb{E}N - \Pr(N \geq 1)\} = \tfrac{1}{4}\{2b\varepsilon - 1 + (1-2\varepsilon)^b\}.$$

Its expansion at zero gives $\mathcal{W}_b / R^*_\kappa = b\varepsilon + O(\varepsilon^2)$.

In the second law, any fitted values in $[0, D]$ at distinct support points satisfy the Lipschitz constraints. At a support point observed $n > 0$ times, the empirical optimum is its target mean, with expected squared residual sum $(n-1)D^2/4$. If O_b is the number of occupied support points, the conditional expected total is $(b - O_b)D^2/4$. Each of the M points is occupied with probability $1 - (1 - 1/M)^b$. Taking expectations gives (12). Independence of Y and U gives the population optimum $D^2/4$, attained by the constant $D/2$. ∎

The second example consists entirely of finite, hence compact, metric spaces. It rules out a vanishing distribution-free additive approximation bound depending only on b and the target range over all such spaces. The support points can be $u_j = De_j/(\kappa\sqrt{2})$ in $\mathbb{R}^M$, where e_j are the coordinate unit vectors. Their diameter is D/κ for every $M \geq 2$, so a bound on representation diameter alone does not remove this limitation. For fixed b, $\mathcal{W}_b / R^*_\kappa = b/(2M) + O(M^{-2})$ as $M \to \infty$. The first example shows why increasing block size can be necessary even on three fixed support points. Larger blocks address these losses, but do not make an uninformative certificate evidence of adequate representation.

4.2 Calibration and certified optimization values

Use disjoint blocks of b independent calibration units, giving $K_b = \lfloor n/b \rfloor$ independent scores. This makes the effect of block size explicit: increasing b tightens $\mathcal{W}_b$ but decreases the number of independently calibrated scores. The hierarchy therefore does not order finite-sample confidence endpoints.

The quadratic program in (7) must be solved from below for a lower certificate. Write its inequalities as $Av \leq c$, where rows of A are coordinate differences and $c_{ij} = \kappa d(U_i, U_j)$. For every nonnegative multiplier vector λ, weak duality gives the lower value

$$d_b(\lambda) = \lambda^\top(AY - c) - \tfrac{1}{4}\|A^\top\lambda\|_2^2 \;\leq\; \min_{Av \leq c} \|Y - v\|_2^2. \tag{13}$$

This follows by minimizing $\|Y - v\|_2^2 + \lambda^\top(Av - c)$ over v; its minimizer is $Y - A^\top\lambda/2$. Thus $\underline{T}_b = \max\{0, d_b(\lambda)/(b-1)\} \leq T_b$. A primal value alone has the wrong inequality direction. The implementation reports primal feasibility and the primal–dual gap; floating-point tolerances are numerical checks rather than proofs by verified arithmetic.

Corollary 6 (Independent-block lower calibration) *Suppose measurable numerical scores $\underline{T}_{b,k} \in [0, B_b^{\mathrm{blk}}]$ satisfy $\underline{T}_{b,k} \leq T_{b,k}$ pointwise, and $K_b \geq 1$. The exact scores $T_{b,k}$ are independent and identically distributed, computed from disjoint iid blocks; their numerical lower approximations may share computation across blocks. For an allocated failure probability $\alpha \in (0,1)$ and $B_b^{\mathrm{blk}} > 0$, let*

$$L_b = B_b^{\mathrm{blk}}\,\mathrm{kl}_-\left(\frac{K_b^{-1}\sum_{k=1}^{K_b}\underline{T}_{b,k}}{B_b^{\mathrm{blk}}}, \frac{\log(1/\alpha)}{K_b}\right), \tag{14}$$

where kl_- *is defined in Appendix B.3. Then* $L_b \le \mathcal{W}_b \le R^*_\kappa$ *with probability at least* $1-\alpha$*. If* $B_b^{\mathrm{blk}} = 0$*, take* $L_b = 0$*. A finite family of block sizes, representations, horizons, and comparison methods can be selected after calibration when the sum of their allocated failure probabilities is at most the declared total.*

Proof Apply (40) to the iid exact scores $T_{b,k} \in [0, B_b^{\mathrm{blk}}]$. Their KL lower endpoint is at most $\mathcal{W}_b$ with failure probability at most α. The map $q \mapsto \mathrm{kl}_-(q, c)$ is nondecreasing for fixed c. Since the numerical sample mean is no larger than the exact sample mean, L_b is no larger than that exact-score endpoint. Theorem 4 then gives $L_b \le \mathcal{W}_b \le R^*_\kappa$. A union bound proves the finite-family assertion; different methods may reuse calibration data. ■

An empirical Bernstein lower bound can instead be applied to the computed scores when $K_b \ge 2$ and each is a common fixed measurable function of its own block, so those scores are iid. The empirical Bernstein endpoint need not be coordinatewise monotone; the preceding KL argument alone does not justify that substitution for coupled numerical outputs. Block sizes and optimization rules are fixed before calibration; taking the largest endpoint across a frozen family requires the stated joint allocation. This is distinct from multiplying the full-sample empirical-minimum endpoint in Appendix B.4 by $n/(n-1)$, which that endpoint's proof does not justify.

Conditioning and forecasting origins. For a common origin or stratum A, draw the b observations independently conditional on A. The same proof gives a monotone hierarchy $\mathcal{W}_{b,A} \le R^*_\kappa$. For finitely many values of A and the compact-support conditions above, its limit is

$$\sum_a \Pr(A = a) \inf_{g \in \mathcal{G}_\kappa(d)} \mathbb{E}[(Y - g(U))^2 \mid A = a],$$

which can be strictly below the optimum of one shared decoder. The decoder attaining each conditional infimum may differ with a. Consequently, averaging a block score over common forecast origins preserves validity and monotonicity but does not generally establish convergence to the shared-decoder risk. For that consistency claim, one may instead draw an independent uniform origin in each independent episode; the resulting observations are independent draws from the full evaluation law. Neither construction treats overlapping windows as independent calibration units.

5 Learning and certifying KAHM forecasts

The procedure has three stages. KAHM reconstruction models map each history to geometric features; a constrained decoder maps those features to future observations; and independent calibration assesses the resulting forecasts. Keeping these stages separate also allows alternative representations to use the same decoder and evaluation procedure.

5.1 Notation and basic conventions

Vectors are columns unless used as data-matrix rows. We write $\|\cdot\|_2$, $\|\cdot\|_{\mathrm{F}}$, and $\|\cdot\|_{\mathrm{op}}$ for Euclidean, Frobenius, and operator norms. Matrix inner products are $\langle M, N\rangle_{\mathrm{F}} = \mathrm{tr}(M^\top N)$. Logarithms are natural; $z_+ = \max\{z, 0\}$; $\|V\|_{L^2} = (\mathbb{E}\|V\|_2^2)^{1/2}$. The projection $\Pi_{\mathcal{Y}}$ onto a nonempty closed convex output set is nonexpansive; for a box it clips coordinates. All random variables and prediction maps are measurable.

Symbol	Meaning
y_t, d_y	Observation at time t and its dimension.
x_t, m	Causal history and number of observed time steps in it.
$\mathcal{H}$, H, q	Set of forecast horizons, its size, and target dimension; for a joint forecast, $q = d_y H$.
Φ_b, C_b	Feature block b and its number of coordinates.
Γ_s, p_s	Stack of the first s feature blocks and total dimension $p_s = \sum_{b=1}^{s} C_b$.
a, W_b	Decoder intercept in $\mathbb{R}^q$ and block coefficients in $\mathbb{R}^{C_b \times q}$; predictions use $W_b^\top \Phi_b$.
κ, λ	Decoder sensitivity budget and ridge regularization parameter.
$R(f)$, R^*	Population mean squared error of a fitted forecast and infimum over a stated decoder class.
$W_{\kappa,A}$	Population pairwise obstruction, using budget κ and conditioning variable A; distinct from coefficient matrices W_b.
T_b, $\mathcal{W}_b$, B_b^{blk}	Scalar block score for b observations, its expectation, and its deterministic range bound; $\mathcal{W}_b$ is distinct from the coefficient matrix W_b.
n, K	Number of calibration episodes and number $K = \lfloor n/2 \rfloor$ of disjoint episode pairs.
$Z_{i,jh}$, $Q_{r,jh}$	Loss of candidate j at horizon h on episode i, and obstruction score on pair r.
L_{jh}, U_{jh}	Lower obstruction endpoint and upper fitted-risk endpoint; the unindexed $U = \Gamma(X)$ instead denotes a random representation.
D_h, B_h	Diameter of the known horizon-h output set and squared-loss bound $B_h = D_h^2$.
$c(X)$, π_a, N_a, K_a	Frozen stratum label, population stratum mass, sample count, and within-stratum pair count.
W_{str}, L_{str}, $\mathcal{P}$	Stratified population witness, its lower endpoint, and the confidence set for stratum masses.
c_j, c_-, c_+, c^*	Declared class cost, lower and upper certified cost endpoints, and minimum feasible cost.
ϵ_h, δ, α	Requested RMSE tolerance, total failure probability, and failure probability allocated to one confidence statement.

Table 2: Recurring notation. Mean squared error (MSE) is an expected squared Euclidean error; root-mean-square error (RMSE) is its square root. Experiment-specific coordinate averaging is stated separately.

Table 2 collects the main quantities. The forecasting construction and the statistical statements below give their precise domains and sampling assumptions.

5.2 A fixed raw-output forecasting task

Let $y_t \in \mathbb{R}^{d_y}$ be the observation at integer time t, where $d_y \geq 1$ is the number of observed coordinates. For a history length $m \geq 1$, define the column vector

$$x_t = (y_{t-m+1}^\top, \ldots, y_t^\top)^\top \in \mathbb{R}^{m d_y}.$$

Covariates available at time t may be appended to this vector. A forecast origin is a time t at which a prediction is made using only this available information. Let $\mathcal{H}$ be a finite, nonempty set of positive integer horizons, ordered increasingly, and let $H = |\mathcal{H}|$ be its size. The joint target is

$$Y_t = (y_{t+h}^\top : h \in \mathcal{H})^\top \in \mathbb{R}^q, \qquad q = d_y H.$$

Thus each training row contains all requested future observations in the same order. The theory also considers one horizon at a time, in which case its target dimension is d_y (or one for a separately scored coordinate).

The history length, any fixed target transformation, horizons, and origin sampling rule define the prediction task. Here "raw output" means the observed signal under that fixed transformation, rather than learned feature coordinates. When comparing history lengths, we use the same origins, starting after the longest history is available. Every candidate is therefore evaluated under the same target and origin law.

The main learner directly predicts each requested horizon, sharing a geometric representation and a joint coefficient budget. It does not generate an entire trajectory by iterating one-step forecasts. The rollout baselines instead iterate a fitted feature-transition model and decode the resulting features. Both approaches are evaluated on the same observed targets; a small one-step feature error alone is not assumed to guarantee an accurate long rollout.

An *episode* is one complete trajectory or other prespecified independent sampling unit. Development data comprise training data, used to fit features and coefficients, and validation data, used to choose hyperparameters or a representative within each model family. Calibration data provide the confidence bounds and may select among candidates frozen before calibration. A separate test set describes forecasting performance after these decisions. Its empirical error estimates population risk; it is not an exact risk value.

5.3 KAHM dictionaries and bounded soft associations

A dictionary block describes a history through its associations with a set of reference groups. KAHM associations depend on local reconstruction models. Radial basis function (RBF) associations instead depend on distances to group centers; affine-hull associations use residuals from fitted local affine subspaces together with a locality term. Each construction specifies which histories receive similar coordinates. A soft association retains a weight for every group, while a hard assignment retains only the selected group. These geometric choices need not preserve all information relevant to forecasting.

For block b, partition the training histories into $C_b \geq 1$ reference groups, indexed by $c = 1, \ldots, C_b$. Let $\mathcal{X}$ be the history domain and let $z(x) \in \mathbb{R}^{d_x}$ be the preprocessed history, with dimension d_x. Each group has $J_{bc} \geq 1$ local KAHM reconstruction models, fitted using its training histories (Kumar et al., 2024, 2025; Kumar and Kargaran, 2026).

Each local model maps a preprocessed query to a reconstruction $\widehat{z}_{bcj}(x)$ using regularized kernel regression on its reference histories. The encoding, covariance-normalized Gaussian kernel, regularizer, and reference histories are all fitted on training data. The reconstruction is an affine combination of local references when its coefficient normalizer is nonzero; the implementation has an explicit convention at a zero normalizer. Supplement Section S1 gives the complete inherited reconstruction equations and numerical rules. The risk results below apply to the fixed map produced by these rules and do not require its coefficients to be nonnegative.

The local subspace cap is part of the representation specification. For an encoder with $N_{\mathrm{loc}} \geq 2$ stored reference histories, let $r_{\max} \geq 1$ be the requested cap. The native construction starts from

$$r_0 = \min\{r_{\max}, d_x, N_{\mathrm{loc}} - 1\}, \qquad r_{\max} = 20 \text{ by default}, \tag{15}$$

and applies the numerical variance and projected-range rules in Supplement Section S1. The stored rank $r \leq r_0$ is reported separately from the cap. The KAHM rule permits full ambient rank; the affine-hull comparator has its own normal-direction convention. A tuned cap must be fixed before independent calibration or included in a declared simultaneously calibrated family. Section 11.2 selects one complete bank on validation and retains its prefixes before calibration.

The folding score combines reconstruction error and angular discrepancy (Kumar and Kargaran, 2026, Section 3.3). Define

$$e_{bcj}(x) = 1 - \exp\{-\|z(x) - \widehat{z}_{bcj}(x)\|_2\}, \qquad \theta_{bcj}(x) = \frac{\arccos \gamma_{bcj}(x)}{\pi},$$

where $\gamma_{bcj}(x)$ is the cosine similarity $\langle z(x), \widehat{z}_{bcj}(x)\rangle / (\|z(x)\|_2 \|\widehat{z}_{bcj}(x)\|_2)$, clipped to $[-1, 1]$; when either vector has zero norm, the implementation sets this similarity to zero. The inverse cosine $\arccos$ takes values in $[0, \pi]$. The group score is

$$T_{bc}(x) = \min_{1 \leq j \leq J_{bc}} \left\{ \frac{e_{bcj}(x)^2 + \theta_{bcj}(x)^2}{2} \right\}^{1/2}.$$

Thus $T_{bc} : \mathcal{X} \to [0, 1]$ is a geometric discrepancy score: smaller values indicate a closer reconstruction match. It is not a class probability. Taking the minimum allows any local model within the group to provide the match.

Fix an exponent $\omega > 0$ and offset $\tau > 0$. The exponent controls the contrast between affinities, and the offset keeps them positive. The reported runs use $\tau = 10^{-6}$. Define

$$\Phi_{bc}(x) = \frac{(1 - T_{bc}(x) + \tau)^{\omega}}{\sum_{r=1}^{C_b} (1 - T_{br}(x) + \tau)^{\omega}}, \qquad \Phi_b(x) = (\Phi_{b1}(x), \ldots, \Phi_{bC_b}(x))^{\top}. \tag{16}$$

Then $\Phi_b(x)$ lies in the probability simplex

$$\Delta_{C_b} = \{v \in \mathbb{R}^{C_b} : v_c \geq 0 \text{ for all } c,\ \mathbf{1}_{C_b}^{\top} v = 1\}.$$

The coordinates express relative reconstruction affinity and need not be calibrated probabilities. We evaluate these weights in log space: with $\ell_{bc} = \omega \log(1 - T_{bc} + \tau)$, subtract $\max_c \ell_{bc}$ before exponentiation and normalization. This leaves the relative affinities unchanged and avoids replacing small summed affinities by uniform weights.

The forecasting method combines these reconstruction features with retained blocks, a constrained output decoder, and independent calibration. In the main experiments, histories are standardized using training means and standard deviations. Section 11.4 separately compares training-based range scaling and bounded raw histories. The input subsample and the maximum number of local training points are fixed by each configuration. Once fitted, the feature maps remain fixed during decoder fitting and calibration.

Why input scaling can change the predictor. Even for a fixed query and reconstruction, multiplying both vectors by a positive scalar a changes the radial term to $1-\exp\{-a\|z-\widehat{z}\|_2\}$, while leaving their angle unchanged. Translating both vectors leaves their difference unchanged but generally changes their angle; unequal coordinate scales can change both terms. Refitting after preprocessing can additionally change the reference groups, retained principal components, and local regularization parameters. Thus a covariance-normalized kernel does not make the complete KAHM representation invariant to input scaling. Training-based scaling to $[-1,1]$ is a model choice whose effect must be evaluated on the forecasting task.

At stage s, retain the first s blocks:

$$\Gamma_s(x) = (\Phi_1(x)^\top, \dots, \Phi_s(x)^\top)^\top \in \prod_{b=1}^{s} \Delta_{C_b} \subset \mathbb{R}^{p_s}, \qquad p_s = \sum_{b=1}^{s} C_b.$$

A *prefix* means these first s blocks. Their cluster partitions need not be nested: preserving earlier forecasts requires retaining the earlier coordinate functions, not their partitions. The experiments append blocks of sizes $2, 4, 8, 16$, giving prefix dimensions $2, 6, 14, 30$.

Connection to the general hierarchy. For fixed learned blocks, the ambient product of simplices is compact under $\rho_s(u,v) = \max_{b\le s} \|u_b - v_b\|_2$, with diameter at most $\sqrt{2}$ independently of s. Thus Theorem 4 applies directly to any bounded scalar observed target and unconditional iid calibration observations from this representation. No continuity of the map from histories to affinities is required. This is an instantiation of the general result; the same conclusion holds for other fixed simplex-valued feature blocks. Increasing the number s of dictionary blocks changes the representation class, whereas increasing the observation-block size in Section 4 strengthens its population lower information while holding that representation fixed.

5.4 A constrained multihorizon decoder

Let (x_i, Y_i), $i = 1, \dots, N$, be the training histories and their joint targets. Form the response matrix $\mathsf{Y} \in \mathbb{R}^{N\times q}$ with row i equal to $Y_i^\top$, and the feature matrix $F = [F_1, \dots, F_s] \in \mathbb{R}^{N\times p_s}$, where row i of $F_b \in \mathbb{R}^{N\times C_b}$ is $\Phi_b(x_i)^\top$. Fit an intercept $a \in \mathbb{R}^q$ and coefficient matrix $W \in \mathbb{R}^{p_s\times q}$ by solving

$$\min_{a,W} \ \frac{1}{N}\|\mathsf{Y} - \mathbf{1}_N a^\top - FW\|_{\mathrm{F}}^2 + \lambda\|W\|_{\mathrm{F}}^2 \quad \text{subject to} \quad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \le \kappa. \tag{17}$$

Here $W_b \in \mathbb{R}^{C_b\times q}$ contains the coefficient rows for block b, $\lambda \ge 0$ is the ridge parameter, and $\kappa \ge 0$ is the coefficient budget. The intercept is unpenalized. The loss sums squared errors over all target coordinates and horizons, then divides by N; it is not additionally divided by q or H.

For a known output box $\mathcal{Y} \subset \mathbb{R}^q$ containing the target, the reported prediction is

$$f_s(x) = \Pi_{\mathcal{Y}}\big(a + W^\top \Gamma_s(x)\big).$$

The objective in (17) fits the unprojected affine predictor; clipping is applied at prediction time and included in every reported loss. Throughout the paper, the same affine map is written as $a + \sum_b W_b^\top \Phi_b(x)$, with $W_b \in \mathbb{R}^{C_b\times q}$.

Centering the training matrices eliminates the intercept. Projected accelerated gradient then solves the constrained quadratic. Each projection computes the block Frobenius norms, projects

that nonnegative vector onto an ℓ_1 ball of radius κ, and rescales the blocks. The recorded first-order optimality gap bounds the remaining training-objective error; a small change between iterates alone would not give that bound. Appendix A derives both the projection and the gap. The fitted-risk certificate remains valid after early stopping, provided the returned model satisfies the budget used for certification.

The ridge term regularizes fitting, whereas the block constraint defines the feasible sensitivity class. Exact block duplication leaves that class unchanged, but distributing coefficients across duplicates can reduce the squared Frobenius ridge penalty. The replication results therefore concern feasible prediction classes and their population obstruction. The duplicate-block experiment uses $\lambda = 0$ to compare these feasible classes without changing the fitting penalty.

5.5 Calibration and model selection

The candidate family includes history lengths, dictionary prefixes, geometric feature types, and baseline models. Each certificate compares a representation lower bound with a fitted-model upper bound for the same observed target, origin law, and decoder sensitivity budget. Algorithm 1 summarizes the procedure; Sections 4 and 6 give the score formulas and their guarantees.

A lower endpoint above tolerance can motivate changing the representation or budget, but does not establish that a particular change will help. Paired episode losses therefore compare old and new fitted forecasts directly. They can certify an improvement even when the representation obstruction is zero. Conversely, the lower bound can exclude a restricted class even if no tested refinement attains the desired accuracy.

In the experiments, cost is the number of retained feature coordinates, with ties resolved by history length and then validation error. This measures representation size. A deployment with unequal feature-evaluation costs can use a different cost function, provided it is fixed before calibration.

6 Calibration and representation refinement

6.1 Independent episodes, dependent observations within episodes

An *episode* is one complete observed trajectory, or another complete sampling unit specified by the application. Let $E_1, \dots, E_n$ be independent and identically distributed episodes, also independent of the data used to construct and freeze the models. For horizon $h \in \mathcal{H}$, let $\mathcal{T}_h$ be a fixed, nonempty set of causal forecast origins, available in every episode and used by every compared candidate. Within an episode, observations may be arbitrarily dependent and their distribution may change with time.

Index the J candidates by $j \in \{1, \dots, J\}$ and write $H = |\mathcal{H}|$ for the number of horizons. Candidate j has a representation Γ_j, a representation pseudometric d_j, and a fixed forecaster $f_{jh} = g_{jh} \circ \Gamma_j$ at horizon h. Its decoder g_{jh} has a known Lipschitz budget κ_{jh}. Let $X_{i,t}$ denote the history at origin t in episode i, and let $y_{i,t+h}$ denote the corresponding single-horizon observation (distinct from the joint target Y_t in Section 5). Define

$$Z_{i,jh} = \frac{1}{|\mathcal{T}_h|} \sum_{t \in \mathcal{T}_h} \|y_{i,t+h} - f_{jh}(X_{i,t})\|_2^2, \qquad R_{jh} = \mathbb{E} Z_{i,jh}. \tag{18}$$

Thus $Z_{i,jh}$ is one average loss per episode, and R_{jh} is its population mean. Equivalently, R_{jh} is the risk for a new episode and an independently, uniformly sampled origin from $\mathcal{T}_h$. It is not a guarantee

Algorithm 1: Learning and certifying KAHM forecasts

1. Specify horizons, causal origins, known target bounds, RMSE tolerances $\epsilon_h \geq 0$, total failure probability $\delta \in (0, 1)$, and a candidate-cost rule.
2. Use development data to fit geometric blocks and constrained decoders for their retained prefixes. Fit comparison models on the same observed targets.
3. Freeze all candidates, preprocessing choices, sensitivity budgets, history-based strata, costs, concentration bounds, and proposed comparisons. Allocate δ across the risk bounds, stratum-mass intervals, alternative partitions, and improvement comparisons.
4. On independent calibration episodes, compute one mean squared loss per episode and horizon. Compute obstruction scores using disjoint episode pairs or frozen-size blocks, using the declared origin sampling rule and the chosen bound's sample-size requirements ($n \geq 4$ for empirical Bernstein pair calibration). For stratified calibration, use one prespecified origin per independent episode (or a justified episode-level stratum), pair within strata, and estimate their masses and confidence intervals.
5. Compute simultaneous lower endpoints L_{jh} and upper endpoints U_{jh} for candidate j at horizon h.
6. Accept a fitted candidate when $U_{jh} \leq \epsilon_h^2$ for every required horizon; exclude its representation class when $L_{jh} > \epsilon_h^2$ for any horizon. Return the least costly accepted candidate, if one exists, and the cost interval between the cheapest nonexcluded class and the cheapest accepted fit.
7. Equal finite cost endpoints certify minimum feasible cost among the declared classes. Exclusion of every class certifies infeasibility within that family. Otherwise report the unresolved accuracy or cost decision explicitly.

Figure 1: Forecasting and certification for a finite family fixed before calibration. The same procedure applies to alternative feature constructions. A candidate fitted after inspecting calibration needs fresh evaluation data or a separately justified adaptive guarantee.

conditional on an individual history. Changing the episode weights or the origin distribution changes the risk being certified.

Assume both targets and predictions lie almost surely in a known nonempty closed convex set $\mathcal{Y}_h$. Let $D_h = \sup_{y,z \in \mathcal{Y}_h} \|y - z\|_2 < \infty$ be its diameter, and put $B_h = D_h^2$. This gives the deterministic loss bound $0 \leq Z_{i,jh} \leq B_h$. Prediction clipping alone does not establish this bound when the target is unbounded. Nor do training minima and maxima establish a population range.

Form $K = \lfloor n/2 \rfloor$ disjoint episode pairs, leaving one episode unused for pairing when n is odd. For $r \in \{1, \dots, K\}$ define

$$Q_{r,jh} = \frac{1}{2|\mathcal{T}_h|} \sum_{t \in \mathcal{T}_h} \Big(\|y_{2r-1,t+h} - y_{2r,t+h}\|_2 - \kappa_{jh} d_j\big(\Gamma_j(X_{2r-1,t}), \Gamma_j(X_{2r,t})\big) \Big)_+^2. \tag{19}$$

For each j, h, the variables $Z_{i,jh}$ are independent across episodes, and $Q_{r,jh}$ are independent across disjoint pairs. Terms within an individual average need not be independent. Their ranges are $[0, B_h]$ and $[0, B_h/2]$, respectively. Write $W_{jh} = \mathbb{E}Q_{r,jh}$. This is exactly the witness in (2) for the horizon-h task when A is a uniformly sampled common origin: conditional on that origin, the two histories and targets come from independent episodes.

For $N \geq 2$ independent, identically distributed scalar observations $V = (V_1, \dots, V_N)$ in a known interval of length $b \geq 0$, define the sample mean and unbiased sample variance by

$$\bar{V} = \frac{1}{N} \sum_{i=1}^{N} V_i, \qquad s_V^2 = \frac{1}{N-1} \sum_{i=1}^{N} (V_i - \bar{V})^2.$$

For a failure probability $\alpha \in (0, 1)$, use the empirical Bernstein radius

$$\mathrm{rad}_{\mathrm{EB}}(V; b, \alpha) = \sqrt{\frac{2 s_V^2 \log(2/\alpha)}{N}} + \frac{7b \log(2/\alpha)}{3(N-1)}. \tag{20}$$

Here $\log$ is the natural logarithm. The bounded-variable inequality of Maurer and Pontil (2009, Theorem 4), also applied to the reflected observations $-V_i$, gives each of the one-sided bounds $\mathbb{E}V_1 \leq \bar{V} + \mathrm{rad}_{\mathrm{EB}}$ and $\mathbb{E}V_1 \geq \bar{V} - \mathrm{rad}_{\mathrm{EB}}$ with failure probability at most α. A joint claim for both directions requires allocation to both events.

Theorem 7 (Simultaneous class and fitted-risk certificate) *Suppose the J candidates and H horizons are frozen using independent development data, $n \geq 4$, and all fitted decoders satisfy their stated Lipschitz budgets. Fix $\delta \in (0, 1)$ and set $\alpha = \delta/(2JH)$. With $\bar{Q}_{jh} = K^{-1} \sum_{r=1}^{K} Q_{r,jh}$ and $\bar{Z}_{jh} = n^{-1} \sum_{i=1}^{n} Z_{i,jh}$, define*

$$L_{jh} = \max\{0, \bar{Q}_{jh} - \mathrm{rad}_{\mathrm{EB}}(Q_{\cdot,jh}; B_h/2, \alpha)\}, \tag{21}$$

$$U_{jh} = \min\{B_h, \bar{Z}_{jh} + \mathrm{rad}_{\mathrm{EB}}(Z_{\cdot,jh}; B_h, \alpha)\}. \tag{22}$$

The dot denotes the full sample over episodes or pairs, so the radius uses sample size K for $Q_{\cdot,jh}$ and n for $Z_{\cdot,jh}$. With probability at least $1-\delta$, conditional on the development data, all candidate–horizon combinations simultaneously satisfy

$$L_{jh} \leq W_{jh} \leq R_{jh}^* \leq R_{jh} \leq U_{jh}. \tag{23}$$

Here R_{jh}^* is the infimum risk over all κ_{jh}-Lipschitz decoders of Γ_j, using d_j, the same horizon, the same uniform origin law, and the same raw target.

Proof Condition on the development data so that the entire candidate family is fixed. For each j, h, apply the upper empirical Bernstein bound to the n episode losses and the lower bound to the K pair scores. Since $0 \le W_{jh}$ and $R_{jh} \le B_h$, truncating the endpoints at zero and B_h preserves validity. A union bound over these $2JH$ one-sided events gives $L_{jh} \le W_{jh}$ and $R_{jh} \le U_{jh}$ jointly, with failure probability at most $2JH\alpha = \delta$.

For any admissible decoder, Theorem 1 applies to the joint law obtained by drawing a common uniform origin and two independent episodes. It bounds W_{jh} by that decoder's risk under the same origin mixture. Taking the infimum over these decoders gives $W_{jh} \le R^*_{jh}$. Finally, the fitted decoder belongs to this class, so $R^*_{jh} \le R_{jh}$. The loss and pair statistics may use the same episodes: independence between the upper and lower statistics is unnecessary for the union bound. ■

The probability $1 - \delta$ concerns repeated draws of the calibration episodes. The theorem bounds expected squared error; it does not bound every future error by $\sqrt{U_{jh}}$ or provide a prediction interval with conditional coverage. Its independent sample sizes are n and K, not the number of overlapping windows. Additional origins may reduce the variance of an episode average, but they do not create additional independent episodes. The representation, normalization, dictionary sizes, budgets, horizons, and all baselines included in the joint guarantee must be fixed before calibration. They may depend on the development data because the theorem conditions on those data.

Stratification under the same target law. A partition can concentrate pairs on distinctions that unconditional pairing rarely encounters. Condition on the development data throughout this construction, so the representation, metric, and budget are fixed. Fix a measurable map $c : \mathcal{X} \to \{1, \dots, M\}$, with $M \ge 1$, using development data only, and write $\pi_a = \Pr\{c(X) = a\}$. For each positive-mass stratum, let W_a be the witness from two independent draws conditional on $c(X) = a$. Set $W_a = 0$ when $\pi_a = 0$. The stratified witness is

$$W_{\mathrm{str}} = \sum_{a=1}^{M} \pi_a W_a \le R^*_\kappa. \tag{24}$$

This is Theorem 1 with $A = c(X)$; weighting by the population masses preserves the original target law.

To estimate it, use $n \ge 1$ independent, identically distributed observations (X_i, Y_i) from that law. For each independent trajectory, draw one origin independently from the prespecified common origin law. This produces independent observations under the episode–origin law. Pooling all dependent windows does not. The fitted-risk upper bound may still average all prescribed origins within each episode; a union bound does not require independence between that bound and the stratified statistic. Let $N_a = \sum_i \mathbf{1}\{c(X_i) = a\}$, where $\mathbf{1}\{\cdot\}$ denotes an indicator, and set $K_a = \lfloor N_a/2 \rfloor$. Set $U_i = \Gamma(X_i)$ and write $(U_{a,k}, Y_{a,k})$, $k = 1, \dots, N_a$, for the observations in stratum a, listed in their original sample-index order. For $r = 1, \dots, K_a$, their disjoint pair scores are

$$Q_{a,r} = \frac{1}{2}\left(\|Y_{a,2r-1} - Y_{a,2r}\|_2 - \kappa d(U_{a,2r-1}, U_{a,2r})\right)_+^2 .$$

Assume the target set has squared diameter $B < \infty$, so $0 \le Q_{a,r} \le B/2$.

Theorem 8 (A stratified lower certificate with estimated masses) *Fix $\delta_p, \beta_1, \dots, \beta_M \in (0,1)$ with $\delta_p + \sum_a \beta_a < 1$. If $K_a < 2$, set $\ell_a = 0$; otherwise set*

$$\ell_a = \max\{0, \bar{Q}_a - \mathrm{rad}_{\mathrm{EB}}(Q_{a,\cdot}; B/2, \beta_a)\}.$$

Here $\bar{Q}_a$ is the mean of the K_a scores. Define

$$t_p = \sqrt{\frac{\log(2M/\delta_p)}{2n}}, \qquad p_a^- = (N_a/n - t_p)_+, \qquad p_a^+ = \min\{1, N_a/n + t_p\},$$

and the confidence set, with $\Delta_M = \{p \in [0,1]^M : \sum_{a=1}^M p_a = 1\}$ the probability simplex, $\mathcal{P} = \{p \in \Delta_M : p_a^- \le p_a \le p_a^+ \text{ for every } a\}$. Then

$$L_{\mathrm{str}} := \min_{p \in \mathcal{P}} \sum_a p_a \ell_a \;\le\; W_{\mathrm{str}} \;\le\; R^*_\kappa \tag{25}$$

with probability at least $1 - \delta_p - \sum_a \beta_a$.

Proof Condition on the complete stratum-label vector. The observations in each stratum are independent draws from its conditional law; their disjoint pair scores are therefore independent and identically distributed. For every realized count with $K_a \ge 2$, the conditional failure probability of $\ell_a \le W_a$ is at most β_a. Sparse strata satisfy it deterministically. Averaging over the labels and taking a union bound gives all score inequalities with failure probability at most $\sum_a \beta_a$; there is no additional union over possible counts. For each stratum, Hoeffding's inequality gives $\Pr\{|N_a/n - \pi_a| > t_p\} \le 2\exp(-2nt_p^2) = \delta_p/M$. A union bound over the M strata therefore places $\pi = (\pi_1, \dots, \pi_M)$ in $\mathcal{P}$ with failure probability at most δ_p. On the intersection of these events, $\min_{p \in \mathcal{P}} \sum_a p_a \ell_a \le \sum_a \pi_a \ell_a \le \sum_a \pi_a W_a$. No independence between masses and score bounds is used. ■

The set $\mathcal{P}$ is nonempty because it contains $(N_a/n)_a$. Its linear minimum is computed by starting at p^- and assigning the remaining mass, up to each upper bound, in increasing order of ℓ_a. Known population masses instead give $L_{\mathrm{str}} = \sum_a \pi_a \ell_a$ with no mass-estimation allocation. For several frozen partitions or candidate–horizon combinations, allocate failure probability over their score events and mass events before taking the largest lower endpoint. A shared partition needs only one mass event. Under the empirical Bernstein convention above, strata with fewer than two pairs contribute no positive evidence; stratification does not increase the number of independent observations.

KL calibration within strata. For $B > 0$ and $K_a \ge 1$, the same theorem holds with

$$\ell_a^{\mathrm{KL}} = \frac{B}{2}\,\mathrm{kl}_-\left(\frac{2\bar{Q}_a}{B}, \frac{\log(1/\beta_a)}{K_a}\right) \tag{26}$$

in place of ℓ_a, and with zero when $K_a = 0$ or $B = 0$. Conditional on the labels, (40) gives each score event with failure probability at most β_a; the same mass event and union bound therefore apply. Returning zero also when $K_a = 1$ is a valid conservative convention. The learned-hierarchy experiment uses that convention and the allocation stated in Section 10. Selecting between KL and empirical Bernstein endpoints after calibration requires allocation to both families.

Proposition 9 (Exactness on compatible strata) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$. Suppose each positive-mass stratum either has a constant representation $U = u_a$ or a joint law supported on two values of (U, Y). In the first case let its optimal prediction be $\mu_a = \mathbb{E}[Y \mid c(X) = a]$; in the second use the fitted values from Proposition 3, with $b = d(u_0, u_1)$ in that construction. If a single global κ-Lipschitz decoder realizes all these values, then $W_{\mathrm{str}} = R^*_\kappa$. For scalar targets on finite support, global compatibility is equivalent to the pairwise Lipschitz inequalities between all the proposed fitted values, including values from different strata.*

Proof A constant representation gives $W_a = \mathbb{E}[\|Y - \mu_a\|_2^2 \mid c(X) = a]$ by the independent-pair variance identity. A two-atom stratum has the finite-support calculation of Proposition 3. The assumed decoder attains their weighted sum, giving equality in (24). For scalar prescribed values v_i at finitely many support points u_i, define $g(u) = \min_i\{v_i + \kappa d(u, u_i)\}$. Each function in this finite minimum is κ-Lipschitz by the triangle inequality for d, and so is their minimum. At u_j, the term $i = j$ equals v_j and every other term is at least v_j by the assumed pairwise inequalities. Thus $g(u_j) = v_j$. Conversely, a global Lipschitz decoder necessarily satisfies all these pairwise inequalities. ■

For the three-atom example, the stratum $\{3\}$ has zero witness. The stratum $\{0, 1\}$ has mass 2ε, and its two atoms are conditionally equally likely, giving conditional witness $1/4$. Consequently, $W_{\mathrm{str}} = 2\varepsilon(1/4) = \varepsilon/2 = R^*_\kappa$. Stratification can thus remove a population gap, rather than merely improve concentration. It need not improve every witness: singleton strata can remove all pairwise conflicts while incompatible predictions across strata still force positive global risk. Conditional optima cannot be combined without the global compatibility condition.

Corollary 10 (Acceptance, exclusion, and inconclusive decisions) *Let $\epsilon_h \geq 0$ be a prespecified root-mean-square error (RMSE) tolerance at horizon h. On the event in* (23)*, $U_{jh} \leq \epsilon_h^2$ certifies that the fitted forecaster attains this tolerance. If $L_{jh} > \epsilon_h^2$, no decoder in the corresponding Lipschitz class attains it. When neither condition holds, the decision is inconclusive. Selecting any member of the frozen family that satisfies $U_{jh} \leq \epsilon_h^2$ at every required horizon preserves these guarantees. In particular, one may minimize a predeclared cost over the certified candidates and abstain if there are none.*

Corollary 11 (A certificate of minimum feasible cost) *Let candidate j represent a class $\mathcal{C}_j$ of multihorizon forecasts through Γ_j, containing its fitted forecast and satisfying its stated horizon-specific Lipschitz budgets. Give it a fixed cost $c_j \geq 0$. Let c^* be the minimum cost of a class containing a forecast that meets every prescribed tolerance. The minimum of an empty set is $+\infty$. For any simultaneously valid lower bounds L_{jh} on the horizon-h risk of every member of $\mathcal{C}_j$, and upper bounds U_{jh} on its fitted forecast, define*

$$\mathcal{E} = \{j : L_{jh} > \epsilon_h^2 \text{ for some } h\}, \qquad \mathcal{A} = \{j : U_{jh} \leq \epsilon_h^2 \text{ for all } h\},$$
$$c_- = \min_{j \notin \mathcal{E}} c_j, \qquad c_+ = \min_{j \in \mathcal{A}} c_j.$$

Thus $\mathcal{E}$ contains the excluded classes and $\mathcal{A}$ contains the accepted fitted candidates. The costs c_- and c_+ refer to the cheapest nonexcluded class and cheapest accepted fit, respectively. On their joint confidence event,

$$c_- \leq c^* \leq c_+. \tag{27}$$

If $c_- = c_+ < \infty$, a least-cost accepted forecast has minimum feasible cost among the specified classes. If every class is excluded, the tolerance requirements are infeasible within their union.

Proof Every accepted fitted forecast meets the tolerances. Every class containing a forecast that meets them is nonexcluded, since one violated horizon lower bound would rule out that forecast. These two inclusions give (27) and both conclusions. A joint coefficient budget can further restrict $\mathcal{C}_j$; exclusion by its larger horizon-wise Lipschitz classes remains valid. ■

If $c_- < c_+$, a cheaper nonexcluded class may still contain an accurate forecast, so selection establishes only the lowest cost among certified candidates. In the experiments, cost counts feature coordinates rather than runtime. Also, (23) implies

$$0 \le \sqrt{R_{jh}} - \sqrt{R^*_{jh}} \le \sqrt{U_{jh}} - \sqrt{L_{jh}}.$$

Thus the endpoint gap bounds the fitted decoder's RMSE suboptimality within the stated class. A large gap does not identify optimization error: the witness and either confidence correction may be loose.

For simple sample-size calculations, one may instead prespecify the Hoeffding radius $b\sqrt{\log(1/\alpha)/(2N)}$. If a fitted risk lies at least $\gamma > 0$ below a squared-error tolerance, then

$$n \ge \frac{2B_h^2 \log(1/\alpha)}{\gamma^2}$$

is sufficient for its Hoeffding upper bound to pass that tolerance with probability at least $1 - \alpha$. If the pair witness exceeds the tolerance by γ, then

$$K > \frac{B_h^2 \log(1/\alpha)}{2\gamma^2}$$

is sufficient for the corresponding lower bound to exclude the tolerance with that probability. For acceptance, the stated sample size makes the radius at most $\gamma/2$. A one-sided Hoeffding event bounds the sample mean by the population mean plus $\gamma/2$, so the upper endpoint is at most the tolerance. For exclusion, the strict sample-size inequality makes the radius strictly smaller than $\gamma/2$. On the event $\bar{Q} \ge W - \gamma/2$, where W is the population pair witness and $\bar{Q}$ is its sample mean, the lower endpoint is therefore strictly above the tolerance. Each event has probability at least $1 - \alpha$. These are sufficient sample sizes, not optimal rates. Exclusion depends on the witness margin, rather than on an unknown oracle-risk margin. Choosing the smaller of separately computed Hoeffding and empirical Bernstein bounds requires confidence allocation to both bounds; otherwise the choice must be fixed in advance.

6.2 Refinement that preserves the meaning of the norm budget

Let $\Phi_b : \mathcal{X} \to \Delta_{C_b}$, $b = 1, \dots, s$, be fixed dictionary blocks, where $\Delta_{C_b} = \{u \in \mathbb{R}^{C_b} : u_c \ge 0,\ \sum_c u_c = 1\}$ is the probability simplex. Retain the entire stack $\Gamma_s = (\Phi_1, \dots, \Phi_s)$. Refinement appends a block while keeping every previous coordinate function unchanged. For stacked vectors $u = (u_1, \dots, u_s)$ and $v = (v_1, \dots, v_s)$ define the block maximum metric $\rho_s(u, v) =$

$\max_{1\le b\le s}\|u_b - v_b\|_2$. Its induced pseudometric on histories and the coefficient norm are

$$d_s(x,x') = \rho_s(\Gamma_s(x),\Gamma_s(x')) = \max_{1\le b\le s}\|\Phi_b(x) - \Phi_b(x')\|_2, \qquad \|W\|_{2,1} = \sum_{b=1}^{s}\|W_b\|_{\mathrm{F}}. \tag{28}$$

Here $\|W\|_{2,1}$ is a sum over *blocks*, not over individual rows. As in (17), $a \in \mathbb{R}^q$ and $W_b \in \mathbb{R}^{C_b\times q}$, so the contribution of block b to a column-vector prediction is $W_b^\top \Phi_b(x)$.

Let $\mathcal{Y} \subseteq \mathbb{R}^q$ be a nonempty closed convex output set containing Y almost surely. Write $\Pi_{\mathcal{Y}}(z) = \arg\min_{y\in\mathcal{Y}}\|z-y\|_2$ for Euclidean projection; for a box, it clips each coordinate to its allowed interval. Define

$$\mathcal{F}_{s,\kappa} = \left\{ x \mapsto \Pi_{\mathcal{Y}}\left(a + \sum_{b=1}^{s} W_b^\top \Phi_b(x)\right) : a \in \mathbb{R}^q,\ \sum_{b=1}^{s}\|W_b\|_{\mathrm{F}} \le \kappa \right\}. \tag{29}$$

The intercept is unconstrained. Group norm constraints are established tools (Bach, 2008); here the block norm makes the decoder's sensitivity compatible with retained dictionary blocks.

Theorem 12 (Nested and replication-invariant refinement) *Every decoder in $\mathcal{F}_{s,\kappa}$ is κ-Lipschitz with respect to d_s. Appending a block gives $\mathcal{F}_{s,\kappa} \subseteq \mathcal{F}_{s+1,\kappa}$ and $d_{s+1} \ge d_s$ pointwise. Consequently, the infimum prediction risk in this decoder class cannot increase. Neither can the pair witness, provided the budget, target law, and conditional pairing law remain fixed. Appending an exact copy of an existing block leaves both the decoder class and the pair witness unchanged.*

Proof For the unprojected decoder, the triangle inequality and the bound $\|Av\|_2 \le \|A\|_{\mathrm{F}}\|v\|_2$ give

$$\left\|\sum_b W_b^\top\big(\Phi_b(x) - \Phi_b(x')\big)\right\|_2 \le \sum_b \|W_b\|_{\mathrm{F}}\|\Phi_b(x) - \Phi_b(x')\|_2 \le \kappa d_s(x,x').$$

Projection onto a closed convex set is nonexpansive, so this bound also holds for the projected decoder. Assigning zero coefficients to a new block embeds each old decoder in the refined class, proving inclusion and hence nonincrease of the optimal risk. Appending a block can only increase the maximum in d_s. For each paired observation, increasing that distance can only decrease the positive-part discrepancy in (2); taking expectations proves witness monotonicity.

If block $s+1$ exactly copies block b, merge its coefficients into the old block. The prediction is unchanged and $\|W_b + W_{s+1}\|_{\mathrm{F}} \le \|W_b\|_{\mathrm{F}} + \|W_{s+1}\|_{\mathrm{F}}$, so the merged coefficients remain feasible. This proves the reverse class inclusion for duplication. A duplicate also leaves the maximum distance unchanged, and therefore leaves the witness unchanged. ■

These claims concern population optima over nested feasible classes, not the risks of separately fitted finite-sample estimators. They also do not assert that every new block contains useful predictive information. Since the Euclidean diameter of each probability simplex is at most $\sqrt{2}$, $d_s \le \sqrt{2}$ regardless of the number of retained blocks. Certifying a fitted decoder at its smaller actual norm is possible, but changes the comparator class. Refinement monotonicity is asserted only at a common budget.

The choice of norm matters even when no information is added. Let two equally likely histories satisfy $\Phi(x_0) = e_1 = (1,0)^\top$ and $\Phi(x_1) = e_2 = (0,1)^\top$, with scalar targets -1 and 1. Under a

single Frobenius constraint of radius one, the largest possible forecast separation is $\sqrt{2}$, and the best affine mean squared error (MSE) is $(2-\sqrt{2})^2/4$. After duplicating the block, the concatenated coefficient vector $(-1/2, 1/2, -1/2, 1/2)$ has norm one and predicts both targets exactly. Duplication has relaxed the effective sensitivity constraint. The block-sum constraint prevents this effect while allowing genuinely new blocks to change the geometry.

Theorem 13 (A necessary amount of geometric change) *Let d_0, d_1 be measurable pseudometrics induced on a common history space by an old and a new representation. Use the same square-integrable target, conditionally independent pairing law, and sensitivity budget $\kappa > 0$. Define the random distance increase $\Delta = (d_1(X, X') - d_0(X, X'))_+$ and assume $\|\Delta\|_{L^2} < \infty$. Let W_0, W_1 be the corresponding witnesses from* (2)*, using these induced history distances. Then*

$$\sqrt{W_1} \geq \left(\sqrt{W_0} - \frac{\kappa}{\sqrt{2}} \|\Delta\|_{L^2} \right)_+ . \tag{30}$$

In particular, if a κ-Lipschitz decoder of the new representation attains RMSE at most $\epsilon \geq 0$, then

$$\|\Delta\|_{L^2} \geq \frac{\sqrt{2}}{\kappa} (\sqrt{W_0} - \epsilon)_+ . \tag{31}$$

On any event $0 \leq L_0 \leq W_0$, replacing W_0 by L_0 on the right side of (31) *is valid simultaneously for every new representation satisfying these conditions. The new representation need not have been fixed when L_0 was computed.*

Proof Set $a = \|Y - Y'\|_2$ and $v_i = (a - \kappa d_i(X, X'))_+$ for $i = 0, 1$. The positive-part function is nondecreasing and 1-Lipschitz. Consequently, $v_0 \leq v_1 + \kappa\Delta$ pointwise, and the L^2 triangle inequality gives

$$\|v_1\|_{L^2} \geq (\|v_0\|_{L^2} - \kappa\|\Delta\|_{L^2})_+ .$$

Since $\|v_i\|_{L^2} = \sqrt{2W_i}$, dividing by $\sqrt{2}$ proves (30). If the new decoder has RMSE at most ϵ, Theorem 1 gives $W_1 \leq \epsilon^2$. Combining the two inequalities and rearranging gives (31). Finally, its right side is nondecreasing in W_0. Replacing W_0 by a nonnegative lower bound therefore weakens the necessary condition, for every admissible new representation on the same event. ■

This theorem rules out a proposed repair when its population distance increase is too small. It is a necessary condition, not a sufficient refinement rule or a guarantee that an estimated distance increase is accurate. The quantity L_0 must lower-bound the old *pair witness*; an arbitrary lower bound on old oracle risk cannot be substituted. If the budget also changes from κ_0 to κ_1, the same proof gives the necessary condition

$$\left\| (\kappa_1 d_1(X, X') - \kappa_0 d_0(X, X'))_+ \right\|_{L^2} \geq \sqrt{2} (\sqrt{W_{\kappa_0,0}} - \epsilon)_+ ,$$

where $W_{\kappa_0,0}$ is the old witness at budget κ_0. This form allows the increased separation to come from a larger budget, a changed representation, or both.

6.3 When an added block can improve prediction

The preceding theorem gives a necessary geometric condition. The next calculation gives a sufficient population condition for one proposed feature block: its cross-moment with the current residual must be nonzero, and the old decoder must have unused norm budget. A positive obstruction alone implies neither condition.

Proposition 14 (Residual correlation and budget slack) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$ and consider the unprojected affine predictor*

$$f_0(x) = a + \sum_{b=1}^{s} W_b^\top \Phi_b(x), \qquad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} < \kappa.$$

Let $\psi : \mathcal{X} \to \mathbb{R}^p$ be a new feature block with $\mathbb{E}\|\psi(X)\|_2^2 < \infty$. This block may take general Euclidean values; it need not be simplex-valued. The same sum of block Frobenius norms constrains its coefficients. Define

$$r = Y - f_0(X), \qquad s_0 = \kappa - \sum_b \|W_b\|_{\mathrm{F}} > 0,$$
$$C = \mathbb{E}[\psi(X) r^\top] \in \mathbb{R}^{p\times q}, \qquad G = \mathbb{E}[\psi(X)\psi(X)^\top] \in \mathbb{R}^{p\times p}.$$

Here C is the feature–residual cross-moment, G is the feature second moment matrix, and s_0 is the unused budget. If $C \neq 0$, let $a_0 = \|C\|_{\mathrm{F}}^2$ and $b_0 = \mathrm{tr}(C^\top G C)$, where tr sums the diagonal entries. Then $b_0 > 0$, and the coefficient block

$$V = \eta C \in \mathbb{R}^{p\times q}, \qquad \eta = \min\{a_0/b_0,\ s_0/\sqrt{a_0}\}$$

is feasible when appended to the old blocks. Its risk improvement is

$$R(f_0) - R(f_0 + V^\top \psi) = 2\eta a_0 - \eta^2 b_0 \geq \eta a_0 > 0. \tag{32}$$

If Y and $f_0(X)$ belong almost surely to a nonempty closed convex set $\mathcal{Y}$, projecting the new forecast onto $\mathcal{Y}$ retains at least this improvement.

Proof The old simplex-valued blocks are bounded, so $f_0(X)$ and hence the residual r have finite second moments. The same assumption on the new block makes C and G well-defined by the Cauchy–Schwarz inequality. Also, $b_0 = \mathbb{E}\|C^\top \psi(X)\|_2^2$. If $b_0 = 0$, then $C^\top \psi(X) = 0$ almost surely, implying

$$\|C\|_{\mathrm{F}}^2 = \langle C, \mathbb{E}[\psi(X) r^\top]\rangle_{\mathrm{F}} = \mathbb{E}[r^\top C^\top \psi(X)] = 0,$$

contrary to $C \neq 0$. Thus $b_0 > 0$ and $\eta > 0$.

For any coefficient matrix $V \in \mathbb{R}^{p\times q}$, expand $\|r - V^\top \psi(X)\|_2^2$ and take expectations to obtain

$$R(f_0 + V^\top \psi) = R(f_0) - 2\langle V, C\rangle_{\mathrm{F}} + \mathrm{tr}(V^\top G V).$$

Substituting $V = \eta C$ gives the equality in (32). Since $\|V\|_{\mathrm{F}} = \eta\sqrt{a_0} \leq s_0$, appending this block respects the total budget. Since $\eta \leq a_0/b_0$, the improvement is at least $\eta a_0 > 0$. Finally, projection onto a closed convex set containing Y cannot increase the new squared error. The old forecast is

already in that set, so its reference risk is unchanged. ■

The condition on the old forecast matters. An improvement between two unprojected affine predictors does not by itself imply an improvement between their separately clipped versions. The proposition also makes no claim when the old decoder exhausts its budget. Its moments are population moments: replacing them with training estimates does not establish the same population improvement. For fitted, clipped models, we therefore compare their episode losses directly.

Fix a horizon h and two forecasts before collecting $n \geq 2$ new independent evaluation episodes. Let Z_i^{old} and Z_i^{new} be their losses on episode i, averaged over the same origins as in (18). Define

$$D_i = Z_i^{\mathrm{old}} - Z_i^{\mathrm{new}} \in [-B_h, B_h], \qquad \bar{D} = \frac{1}{n}\sum_{i=1}^{n} D_i, \qquad R^{\mathrm{old}} = \mathbb{E}Z_i^{\mathrm{old}}, \quad R^{\mathrm{new}} = \mathbb{E}Z_i^{\mathrm{new}}.$$

For $\alpha \in (0,1)$, the lower empirical Bernstein bound gives, with probability at least $1-\alpha$,

$$R^{\mathrm{old}} - R^{\mathrm{new}} \geq \bar{D} - \mathrm{rad}_{\mathrm{EB}}(D; 2B_h, \alpha). \tag{33}$$

The interval length is $2B_h$, because D_i can be negative. A positive right side certifies a risk reduction for these two fitted predictors; a nonpositive value is inconclusive. For multiple fixed comparisons, allocate failure probability over the entire comparison family.

Sequential refinement can also be adaptive. At stage s, a candidate may depend on all previous results. Conditional on this full past, its evaluation episodes must be new independent draws from the specified evaluation law. Choose stage failure probabilities $\alpha_s > 0$ in advance, with $\sum_s \alpha_s \leq \delta$, and accept a change only when its paired lower bound is positive. Conditional validity at each stage and a union bound then make every accepted improvement valid on one event of probability at least $1-\delta$. This argument does not permit repeated tuning on a fixed calibration set.

Finally, a positive obstruction need not be removable by refinement. If Y is independent of X and takes values $-a$ and a with equal probability, where $a > 0$, then the minimum risk over all predictors based on X is a^2. Under unconditioned independent pairing, a constant representation has pair witness a^2; no representation of the same observed history removes this independent future noise. Conversely, a zero witness need not imply a useful predictor. For example, suppose every episode follows the same deterministic trajectory, but its target changes across forecast origins. Pairing two episodes at the same origin gives identical targets and hence a zero witness. A constant representation still cannot predict the variation across origins; its optimal risk is $\mathbb{E}\|Y - \mathbb{E}Y\|_2^2$ under the origin law. Obstruction, fitted risk, and inconclusive decisions therefore answer distinct questions and should be reported separately.

7 Population improvement versus calibration precision

The block hierarchy removes the pair witness's fixed population gap under the assumptions of Theorem 4. It does not imply that larger blocks always give a more useful certificate at a fixed sample size. We separate these questions with the exact examples in Proposition 5 and all eight finite-support laws in the supplement's complete finite-support comparison. The laws were examined in the earlier pairwise comparison; the new protocol freezes the complete block-size grid and fresh random streams before drawing its observations.

Table 3: Mean lower endpoints at $n = 4096$ over all 100 repetitions of each law. R^* is the minimum risk in the same scalar Lipschitz class. Columns L_b use disjoint blocks of size b; ERM uses the full sample. A single failure budget 0.05 per law and repetition covers all six methods and all three sample sizes. The hierarchy orders population witnesses, not these finite-sample endpoints.

Law	R^*	L_2	L_4	L_8	L_{16}	L_{32}	ERM
Collapsed binary	1.0000	0.9229	0.9355	0.9375	0.9311	0.9159	0.9089
Two-atom partial	0.3315	0.2776	0.2726	0.2573	0.2330	0.1995	0.2755
Three-point V	0.2222	0.1769	0.1719	0.1589	0.1393	0.1131	0.1763
Five-point zigzag	0.5400	0.3247	0.3870	0.4203	0.4158	0.3831	0.4697
Partial collapse	0.2533	0.0590	0.0964	0.1276	0.1373	0.1244	0.2044
Noisy affine	0.2500	0.0678	0.1017	0.1280	0.1352	0.1219	0.2012
Rare conflict	0.0250	0.0001	0.0004	0.0007	0.0008	0.0005	0.0116
Affine mismatch	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Protocol. For every law, 100 independent streams supply nested samples of $n \in \{256, 1024, 4096\}$ observations. For each sample, we compare disjoint blocks of sizes $b \in \{2, 4, 8, 16, 32\}$ and the full-sample empirical-minimum bound in Appendix B.4. All methods concern the same scalar 1-Lipschitz class. Block calibration uses the sharp target-range bound B_b^{blk} for $Y \in [-1, 1]$ and the KL endpoint in (14). The full-sample comparator uses loss bound four. Each of the 18 method–sample-size endpoints receives failure probability $0.05/18$. Their simultaneous statement is per law and repetition; it is not a joint claim over all 800 streams.

At size two we evaluate the pair formula. Other block scores and the full-sample comparator use feasible dual lower values. Every one of the 45,579 distinct empirical programs has primal–dual gap at most 4.01×10^{-11} in empirical-MSE units. Repeated count vectors reuse the identical optimization result. This finite-support implementation is not a timing comparison for large continuous representations. All raw categorical draws, dual certificates, and 14,400 endpoints are retained.

Results and interpretation. For partial collapse at $n = 4096$, the mean endpoint increases from 0.0590 for pairs to 0.1373 at $b = 16$, then decreases to 0.1244 at $b = 32$. The mean uncalibrated scores continue increasing, from 0.0864 to 0.2195 to 0.2370. The decrease in the endpoint therefore reflects calibration uncertainty, not a reversal of the population hierarchy. Full-sample empirical minimization remains stronger on this law, with mean endpoint 0.2044. It is also stronger on noisy affine, five-point zigzag, and rare-conflict laws. Larger blocks are an intermediate statistical and computational choice, not a uniformly superior replacement for full-sample optimization.

When the pair witness is already exact, block growth can be unnecessary. On the partially separated two-atom law, the mean endpoint decreases from 0.2776 for pairs to 0.1995 at $b = 32$. By contrast, collapsed binary noise has a constant population witness but benefits slightly from the smaller score range and variability of moderate blocks. The affine-mismatch law remains correctly uninformative for every method because its full Lipschitz optimum is zero. These different outcomes are all shown in Table 3.

There are no observed simultaneous coverage violations in the 800 law–repetition instances. This is a numerical check against known optima, not a proof or a replacement for the confidence theorem. The exact curves in Figure 2 separately show why block size must increase to capture

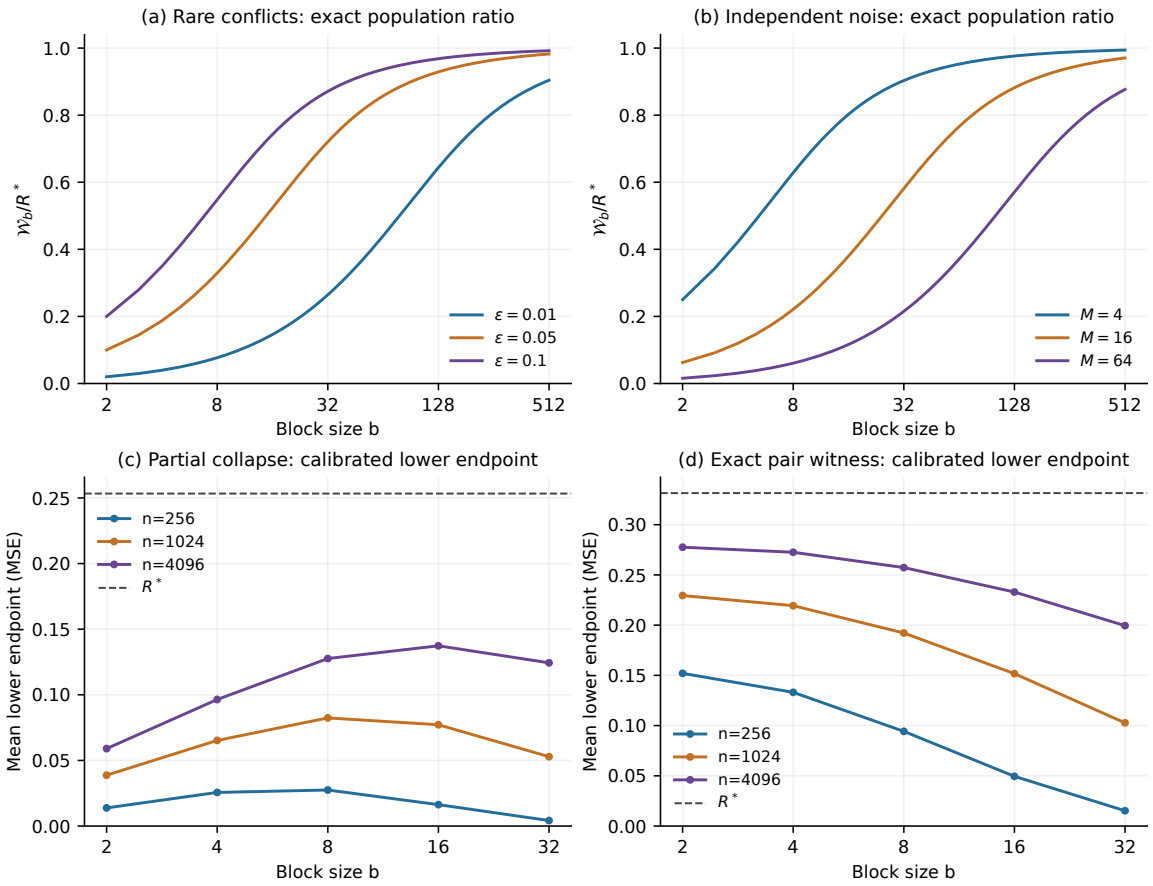


Figure 2: Population and finite-sample effects of block size. Top: exact ratios from Proposition 5, with no Monte Carlo approximation. Bottom: mean calibrated endpoints over every prescribed repetition for two contrasting laws; dashed lines are their population optima. The monotone population hierarchy does not impose monotonicity on the calibrated endpoints. Table 3 reports all eight laws, including cases favoring the full-sample comparator.

sufficiently rare conflicts or noise on many distinct inputs. Finite-sample block size should be fixed using development evidence, or selected from a family covered by a joint confidence allocation.

8 Complementary lower-bound and decision checks

The block study extends an earlier comparison of unconditional pairs, range-aware centered bounds, certified empirical minima, and estimated-mass stratified pairs. That comparison uses the same eight scalar finite-support laws, 300 repetitions at each $n \in \{64, 256, 1024, 4096\}$, and separate 0.05 one-sided allocations per method; it does not assert one joint event across methods. The supplement retains every population optimum, endpoint, tolerance decision, and timing result.

On partial collapse at $n = 4096$, stratification raises the population witness from 0.0867 to the exact risk 0.2533. Mean calibrated endpoints are 0.0646 for unconditional pairs, 0.1772 for

estimated-mass stratification, and 0.2178 for empirical minimization. Finer conditioning is not uniformly better: the chosen partition lowers the three-point population witness from 0.2222 to 0.1667. Rare conflicts remain difficult. At $n = 4096$, the exact MSE is 0.025, yet neither primary pair method excludes RMSE 0.10 in any of 300 trials, whereas empirical minimization does so in every trial. Supplying a tighter known score range raises stratified exclusions to 284; also supplying known masses raises them to 300. The result depends on both the pairing construction and the structural information provided.

All four methods have zero observed coverage violations in this prescribed grid, which is a numerical check rather than a proof. Finite-support lookup and repeated-count optimization make its computation times unsuitable as continuous-representation scalability evidence. A separate two-atom calibration check has one pair-lower violation in 28,800 trials and none for the centered lower or fitted upper bound. Accounting for known zero representation variance substantially strengthens centered calibration in the collapsed case. Thus these experiments do not establish uniform superiority of pairwise certificates over centered or empirical-minimum alternatives.

8.1 An analytic check of the cost decision

An explicit-feature control uses independent histories $X = (Z, S)$ and targets $Y = 0.8SZ + 0.1N$ (repairable) or $Y = 0.8SR + 0.1N$ (irreducible), with $S \sim \mathrm{Bernoulli}(0.2)$ and independent random signs Z, R, N. Its three retained classes have costs $1, 3, 5$ and exact minimum MSEs $0.138, 0.138, 0.01$ in the repairable regime, and 0.138 for all three in the control. These are explicit maps, not learned KAHMs. At RMSE tolerance 0.30, upper calibration accepts the cost-5 fit in every one of 100 repairable repetitions from $n = 1024$. Stratified lower calibration certifies its minimum feasible cost in every repetition from $n = 16384$; unconditional calibration never closes the cost interval through $n = 65536$. Both selection rules return the same fitted model: the additional conclusion is necessity of its declared cost. The irreducible control is jointly excluded in every repetition only at $n = 65536$ under stratification, with smaller samples and unconditional bounds retained as failures. The supplement gives the complete design, all intermediate outcomes, and simultaneous allocation.

9 Certified decisions for learned KAHM representations

We ask whether class-level decisions can arise when the representation and the pairing partitions are learned by the actual KAHM implementation. Section 8.1 separately examines explicit feature maps. This is a constructed mechanism experiment with continuous histories and fresh independent episodes. It tests the connection between the theory and learned geometry; it does not estimate how often the method succeeds on natural forecasting tasks.

A smooth two-lag product and an irreducible control. Each history is $X = (X_1, X_2)$, where independently $X_j = S_j(0.9 + V_j)$, S_j is uniform on $\{-1, 1\}$, and $V_j \sim \mathrm{Unif}[-0.1, 0.1]$. The repairable one-step target is

$$Y = 0.8X_1X_2 + E, \qquad E \sim \mathrm{Unif}[-0.05, 0.05],$$

with independent noise. This is a smooth nonlinear second-order recurrence at the declared forecast origin, with four modes in the initial-history law. In the irreducible control, an independent history $\widetilde{X}$, unavailable to the predictor, replaces X in the target formula. In both regimes $Y \in [-0.85, 0.85]$.

We fix the MSE tolerance at 0.10 in these target units, after five development-only pilot seeds and before any confirmatory calibration data. It is a study-design tolerance, not an external application requirement. The noise variance is $0.05^2/3$; the irreducible control has Bayes risk $0.8^2(0.9^2 + 0.1^2/3)^2 + 0.05^2/3$. Neither reference value enters model selection or a reported confidence endpoint.

Learned classes and frozen calibration. Each of 20 independently generated trials per regime uses 2,048 training, 1,024 validation, and 16,384 calibration episodes, with one origin and one future observation per episode. Training-only coordinate min–max scaling, as defined in Section 11.4, precedes geometry fitting; frozen extrema transform later histories without clipping. KAHM blocks have two and four clusters, association exponent 16, retained subspace dimension 2, and at most 48 local references per cluster. The first class uses the two-coordinate block; the second retains it and appends the four-coordinate block. Thus their declared costs are 2 and 6. Both use the common sum-of-block-norms budget $\kappa = 2$ and the corresponding maximum-block distance. The cost comparison concerns these two classes and counts retained coordinates, not runtime or all possible forecasting methods.

For each class, validation selects among ridge penalties 0, 10^{-5}, and 10^{-3}. All 40 feature banks, 240 candidate fits, and 80 selected heads are recorded, and every selection is frozen before any calibration episode is generated. Twenty-two candidate fits, including nine selected fits, reach the 20,000-iteration limit before the strict 10^{-9} objective-gap criterion; their largest gaps are 5.35×10^{-9} and 2.21×10^{-9}, respectively. All coefficient budgets remain feasible. The risk guarantees require fixed feasible forecasts, not exact optimization.

Two history-only partitions take the argmax of the frozen two-cluster or four-cluster association block. No generated quadrant labels or future targets define these partitions. We compute unconditional and both stratified lower certificates, accounting for estimated stratum masses, at calibration sizes 256, 1,024, 4,096, and 16,384. A total failure allocation of 0.05 covers *both regimes, all trials, all sample-size prefixes, both classes, and all reported bounds jointly*: 0.01 for upper endpoints, 0.01 for unconditional lower endpoints, and 0.03 for the two stratified families. The latter split each partition's allocation equally between mass estimation and conditional mean bounds. All designs, sample sizes, tolerances, and partitions remain fixed after calibration starts.

What the lower information changes. At 1,024 episodes, upper calibration accepts the six-coordinate forecast in all 20 repairable trials. Stratified lower calibration additionally excludes the cheaper class in 17 trials, certifying minimum feasible cost six; the unconditional lower certificate does so in none. At 4,096 episodes both lower constructions establish this conclusion in all 20 trials. In the irreducible control, stratification excludes both classes in all 20 trials at 4,096 and 16,384 episodes, whereas unconditional lower calibration never establishes joint infeasibility at the prescribed sample sizes. Every trial at 256 episodes remains unresolved. Table 4 retains these small-sample failures.

At 16,384 episodes, the repairable six-coordinate forecast has mean empirical MSE 0.00431 and mean upper endpoint 0.00906 across trials. Its cheaper class has stratified lower endpoints between 0.2513 and 0.3905. In the irreducible control, the six-coordinate class has stratified lower endpoints between 0.3441 and 0.3657, while its unconditional lower endpoints remain between 0.0808 and 0.0929, below the tolerance. The choice of conditioning still matters: the four-cluster partition yields zero lower endpoints for the cheaper class in the repairable regime because it removes the relevant target contrast. The prespecified two-cluster partition retains that contrast. Thus stratification has decision value here, but finer conditioning is not uniformly preferable. These outcomes supply

Regime	Episodes	Upper accepts	Unconditional	Stratified
Repairable	256	0/20	0/20	0/20
Repairable	1,024	20/20	0/20	17/20
Repairable	4,096	20/20	20/20	20/20
Repairable	16,384	20/20	20/20	20/20
Irreducible	256	0/20	0/20	0/20
Irreducible	1,024	0/20	0/20	0/20
Irreducible	4,096	0/20	0/20	20/20
Irreducible	16,384	0/20	0/20	20/20

Table 4: Decisions for fitted KAHM representations over 20 fresh trials per regime. The two retained classes have costs 2 and 6, common budget $\kappa = 2$, and fixed MSE tolerance 0.10. Upper accepts counts trials in which a fitted model passes. The last two columns count certification of minimum cost 6 in the repairable regime, or exclusion of both classes in the irreducible regime. Stratification takes the maximum of separately budgeted lower bounds from two frozen learned partitions. All bounds, trials, regimes, and sample-size prefixes share a total failure allocation of 0.05.

a learned-KAHM example of the class-exclusion mechanism; the natural-data results and their unresolved lower certificates retain their separate scope.

10 The block hierarchy on learned KAHM geometry

The preceding learned-representation study evaluates pair certificates. We now apply the larger-block hierarchy to the same collection of fitted geometries, using fresh independent calibration episodes. This directly tests whether joint consistency among more than two observations changes a class-level decision on learned KAHM features.

Frozen representations and new observations. All 40 archived feature banks and 80 validation-selected decoders from Section 9 are retained, with no refitting or selection of favorable banks. The observed target laws, training-fitted scalers, maximum-block metric, budget $\kappa = 2$, costs $2, 6$, and MSE tolerance 0.10 are unchanged. The new numeric seed rule and complete configuration are frozen before drawing any new calibration episodes. Each bank receives 4096 independent observations, with nested prefixes $n \in \{256, 1024, 4096\}$. The conclusions are conditional on this fixed bank collection; these are not new independent training repetitions.

At each sample size, we compare disjoint observation blocks of $b \in \{2, 4, 8, 16\}$, both learned history partitions, and the fitted-model upper bound. Full-sample empirical-minimum calibration uses all observations at $n = 256$; it is not computed at the two larger sizes. Every lower method uses the same scalar Lipschitz class. The block and stratified mean bounds use KL calibration; the upper endpoints use empirical Bernstein. The 1760 endpoints share one failure probability 0.05, with equal allocation $\eta = 0.05/1760$ per endpoint. For a partition with M strata, its stratified endpoint allocates $\eta/2$ to the joint mass event and $\eta/(2M)$ to each conditional KL mean in (26). Strata with fewer than two pairs receive lower endpoint zero, a conservative convention also used by the recorded implementation. Taking maxima over the frozen methods is covered by this event. The

Regime	Episodes	Upper	$b = 2$	$b = 4$	$b = 8$	$b = 16$	Strat.	ERM
Repairable	256	0/20	0/20	0/20	0/20	0/20	0/20	0/20
Repairable	1,024	20/20	20/20	20/20	20/20	20/20	20/20	–
Repairable	4,096	20/20	20/20	20/20	20/20	20/20	20/20	–
Irreducible	256	0/20	0/20	0/20	0/20	0/20	0/20	20/20
Irreducible	1,024	0/20	0/20	0/20	20/20	20/20	20/20	–
Irreducible	4,096	0/20	0/20	20/20	20/20	20/20	20/20	–

Table 5: Fresh calibration conditional on all 40 archived learned KAHM banks. Upper counts accepted fitted forecasts; lower-method columns count minimum cost six in the repairable regime and exclusion of both classes in the irreducible regime. All 1,760 endpoints share total failure probability 0.05. Strat. takes the maximum of the two separately allocated learned-partition bounds. Full-sample ERM is evaluated only at $n = 256$, on the same observations as the blocks; dashes mean uncomputed. Counts describe the fixed bank collection, not independent new training repetitions.

earlier learned-pair study used empirical Bernstein lower means, so cross-study changes cannot be attributed to block size alone.

What larger blocks add. In the independent-future control at $n = 1024$, blocks of sizes eight and sixteen exclude both classes for all 20 banks, whereas pairs and size-four blocks do so for none. For the six-coordinate class, mean lower endpoints are 0.04575, 0.08829, 0.13100, and 0.14615 as block size increases; the last two cross the tolerance. At $n = 4096$, size-four blocks also exclude both classes for every bank, while pairs still do not. This supplies a learned-geometry example in which the hierarchy changes a decision under matched calibration. The obstruction here includes independent future noise; it is not evidence that a better KAHM representation could remove that noise.

The stronger alternatives and unfavorable cases are retained. Estimated-mass stratification excludes both classes in all 20 control banks at $n = 1024$ and 4096, and its mean six-coordinate endpoint is larger than any tested block endpoint. Full-sample empirical minimization already excludes both classes for every control bank at $n = 256$, where none of the block methods resolves joint infeasibility. In the repairable regime, all block sizes, including pairs, certify minimum cost six for all banks at $n = 1024$ and 4096; larger blocks add no cost decision there. No fitted forecast passes at $n = 256$. Larger blocks also need not give a larger finite-sample endpoint: for the repairable cost-two class at $n = 1024$, the mean falls from 0.20819 at $b = 8$ to 0.18430 at $b = 16$.

Numerical and reproducibility scope. The signed-edge solver in Appendix A.1 retains feasible dual lower values even at the iteration limit. Across 143440 programs, 379 miss the prescribed normalized gap threshold 10^{-7}; all remain in the results. The largest recorded gap is 6.41×10^{-7} and the largest primal feasibility residual is 2.18×10^{-16}. These are floating-point checks, not interval-arithmetic proofs. Complete observations, signed duals, primal vectors, endpoints, and decisions are archived. Independent replay without project imports reproduces all 1760 endpoints and 880 decisions; the largest endpoint discrepancy is 1.01×10^{-12}.

A final provenance check detected late writes to an auxiliary development benchmark log by a concurrent development process. The frozen log and its later version are both preserved. Every executable dependency, protocol, frozen model, and confirmatory numerical record was unchanged; final summary generation used the already completed records without resampling. Timings are retained as execution diagnostics and may include that contention. They do not establish a runtime advantage, and the full-sample comparator's limited sample-size grid does not establish scalability beyond $n = 256$.

11 Representation behavior on broader forecasting tasks

The controlled studies separate population obstruction, calibration uncertainty, and decisions between declared classes. The following experiments examine whether the implemented representations also give useful forecasts on broader tasks. They do not establish universal KAHM superiority or the prevalence of informative natural-data class exclusion. All prescribed configurations and adverse outcomes are retained. The accompanying supplement gives the complete forecasting, local-dimension, preprocessing, matched-geometry, and speech tables and protocols; Appendix C defines the generators and comparison models. The fixed-configuration forecasting, preprocessing, matched-geometry, and natural-data comparisons use local KAHM cap $r_{\max} = 2$. Section 11.2 evaluates dimension selection separately. These caps constrain each local reconstruction model; they are distinct from the retained feature counts $2, 6, 14, 30$.

11.1 Fixed-cap forecasting and retained-block refinement

The fixed-cap reference study uses four bounded length-72 episode laws: harmonic, quasiperiodic, nonlinear autoregressive, and logistic. Five fits per law use 128 training, 512 validation, 4096 calibration, and 2048 test episodes; histories have lengths 4 or 12 and horizons are 1, 4, and 12. Forty candidates per fit are frozen before calibration. Van der Pol and Duffing supply three additional fits each. Calibration averages over origins within independent episodes. Its failure allocation is per declared repetition, rather than joint over the whole research campaign.

KAHM improves on long-horizon persistence for periodic signals, but quadratic and autoregressive regressors are stronger. For nonlinear AR at horizon 12, KAHM RMSE is 0.0743, versus 0.0802 for ridge AR and 0.0813 for quadratic regression; those alternatives are better at one step. At all-horizon RMSE tolerance 0.35, its 800 candidates give 301 acceptances, 24 exclusions, and 475 unresolved decisions. For the 240 oscillator candidates, the corresponding counts are 55, 30, and 155. Van der Pol horizon-12 KAHM RMSE is 0.3814, versus 0.2990 for affine-hull blocks; on Duffing, KAHM has 0.2499, versus 0.1369 for quadratic regression. Direct fitting also does not uniformly improve on rollout: iterated KAHM gives 0.3408 on Van der Pol.

Among 480 four-law retained-KAHM candidate–horizon statements, 119 lower endpoints are positive, but only two exclude tolerance 0.35 and none excludes 0.50. Paired fitted-loss comparisons certify 72 of 360 main, 43 of 54 Van der Pol, and 10 of 54 Duffing adjacent-prefix improvements. These fitted improvements differ from class exclusion. Across all 62 four-law, oscillator, and budget-sensitivity fits, all 186 lower endpoints of the complete-family validation winners are zero; 184 of 186 are zero when selection is restricted to KAHM. Fresh budget comparisons at $\kappa \in \{1, 4, 16\}$ show benefits from increasing one to four on several tasks and little further gain at sixteen. Zero-ridge duplicate-block checks agree within solver gaps; ridge-regularized fits need not be invariant.

Law	Cap 2	Native 20	Selected cap	Change (%)
Harmonic	0.051868	0.051727	0.048401	-6.68
Quasiperiodic	0.037484	0.038452	0.037484	+0.00
Nonlinear AR	0.004591	0.004534	0.004534	-1.24
Logistic	0.164773	0.167156	0.163252	-0.92
Van der Pol	0.060314	0.058421	0.052366	-13.18
Duffing	0.028750	0.034536	0.027932	-2.85

Table 6: Local-dimension policies for the block-direct KAHM decoder. Entries are test MSE averaged over horizons within each law and then over five fitting repetitions for the first four laws and three for each oscillator. The last column compares selected cap with cap two. Validation selects the history and prefix within each policy; the selected-cap arm also selects the cap. These test-error contrasts are descriptive. Supplement Section S9 reports per-horizon errors, variation across fits, comparators, and the simultaneous risk statements.

11.2 Local dimension selection and certificate utility

Local PCA dimension affects the reconstruction map and its induced decoder class. A separate study compares fixed cap two, native cap twenty, and validation selection over $\{1, 2, 4, 8, 12, 20\}$ under the block-direct decoder. It uses 26 fresh fitting repetitions on the same six synthetic laws: five for each of the four reference laws and three per oscillator, with 128/512/4096/2048 training/validation/calibration/test episodes. Inputs are training-standardized; histories, horizons, reference limits, $\kappa = 4$, and decoder ridge 10^{-4} follow the reference protocol.

Each policy selects one history–cap bank by its best prefix's mean validation MSE over horizons, then freezes all four prefixes of that bank. This preserves their nested classes. RBF and affine-hull controls retain their original definitions; autoregressive, quadratic, persistence, and constant forecasts complete the comparison. All 1820 development fits finish, and every retained model and validation choice is sealed, before any calibration episode is generated. The 624 evaluated model instances yield 3744 risk endpoints and 936 planned improvement endpoints, covered jointly by one failure probability 0.05. Test generation follows the complete calibration seal. Supplement Section S9 gives the full protocol and allocation; this event concerns the dimension study alone.

Validation-selected cap reduces mean test MSE on five laws and reproduces cap two on quasiperiodic data (Table 6). Across repetitions, it improves in 18, ties in 8, and worsens in none. It selects cap one, two, or four in 8, 8, and 10 repetitions, respectively. All 78 selected KAHM forecasts use 30 retained coordinates. Native cap twenty has mixed effects, including a 20.12% increase in mean Duffing MSE. Its realized local ranks are three, four, or twelve. Selected KAHM has lower mean MSE than the non-KAHM comparators on nonlinear AR and logistic; quadratic regression remains stronger on the other four laws.

The forecasting gains do not change the selected-forecast tolerance decisions. For every policy, 5, 13, and 21 of the 26 selected forecasts are accepted at RMSE tolerances 0.2, 0.35, and 0.5; all remaining decisions are unresolved. Decisions agree across policies for every repetition and tolerance, and all 234 selected-KAHM horizon lower endpoints are zero. There are 132 positive improvement lower endpoints among 702 adjacent-prefix comparisons, but none among the 234 selected-policy comparisons. Thus dimension selection improves descriptive forecasting accuracy here without

		MSE $\times 10^3$: mean (SD)			KAHM change (wins)	
Task	Head	KAHM	RBF	Affine	vs. RBF	vs. Affine
Harmonic	Block	49.77 (1.33)	50.97 (0.98)	52.45 (0.78)	-2.4% (5)	-5.1% (5)
	Frobenius	47.67 (1.31)	48.41 (1.13)	44.94 (1.13)	-1.5% (5)	+6.1% (0)
	Rollout	52.26 (1.04)	52.38 (0.98)	51.66 (1.42)	-0.2% (3)	+1.2% (1)
Quasiperiodic	Block	34.26 (1.32)	34.95 (0.70)	34.74 (2.82)	-2.0% (2)	-1.4% (2)
	Frobenius	33.72 (1.30)	34.30 (0.61)	25.92 (2.72)	-1.7% (3)	+30.1% (0)
	Rollout	39.22 (3.18)	41.75 (3.83)	37.95 (4.58)	-6.1% (2)	+3.3% (1)
Nonlinear AR	Block	4.56 (0.05)	4.69 (0.04)	4.77 (0.04)	-2.7% (5)	-4.4% (5)
	Frobenius	4.56 (0.05)	4.69 (0.04)	4.77 (0.04)	-2.7% (5)	-4.4% (5)
	Rollout	4.57 (0.08)	4.81 (0.07)	4.81 (0.05)	-5.0% (5)	-4.9% (5)
Logistic	Block	165.48 (4.28)	167.18 (2.74)	173.27 (2.22)	-1.0% (3)	-4.5% (5)
	Frobenius	160.66 (5.06)	161.29 (3.38)	170.63 (2.33)	-0.4% (2)	-5.8% (5)
	Rollout	167.63 (5.93)	164.90 (5.64)	177.74 (2.40)	+1.7% (1)	-5.7% (5)
Van der Pol	Block	52.92 (5.56)	61.82 (3.52)	48.35 (3.07)	-14.4% (5)	+9.5% (2)
	Frobenius	26.42 (4.55)	30.92 (7.17)	13.03 (2.92)	-14.5% (3)	+102.7% (0)
	Rollout	33.36 (4.97)	40.23 (16.47)	17.30 (3.63)	-17.1% (4)	+92.8% (0)
Duffing	Block	24.82 (1.60)	23.91 (2.21)	21.95 (2.32)	+3.8% (3)	+13.1% (1)
	Frobenius	23.41 (1.76)	21.21 (2.77)	13.72 (1.51)	+10.4% (0)	+70.6% (0)
	Rollout	27.41 (3.03)	27.02 (4.58)	15.23 (1.41)	+1.4% (3)	+79.9% (0)
PenDigits	Block	5.14 (0.32)	6.09 (0.22)	5.89 (1.15)	-15.6% (5)	-12.7% (4)
	Frobenius	5.04 (0.32)	6.09 (0.22)	5.14 (0.70)	-17.2% (5)	-1.9% (4)
	Rollout	5.46 (0.40)	6.65 (0.34)	5.54 (0.97)	-17.9% (5)	-1.4% (3)

Table 7: Matched geometry comparison after validation-only selection from identical preprocessing, history, and dimension options. Absolute all-horizon test MSE and its sample standard deviation across five fitting repetitions are scaled by 10^3. Relative changes use ratios of mean MSEs; negative values favor KAHM. Wins count repetitions with lower KAHM MSE out of five. PenDigits averages both coordinates and weights the same 13 writers equally. Every effect is descriptive; calibrated comparisons are reported separately.

establishing a simultaneously certified policy improvement or increased selected-forecast decision resolution.

11.3 Matched representation and decoder comparisons

KAHM, RBF, and affine-hull geometries each receive three decoders: direct block-norm fitting, direct Frobenius-norm fitting, and feature rollout. Each geometry–decoder pair has the same prefix dimensions $2, 6, 14, 30$, two history lengths, and four preprocessing choices. Validation alone selects among these options. Five new seeds generate fresh synthetic episodes; PenDigits uses new fitting randomness on its existing writer cohort. Equal numerical budgets and feature counts do not imply equal statistical capacity or computation cost.

All 13,440 fits completed. KAHM has lower mean MSE in 28 of 42 contrasts and higher MSE in 14. These are descriptive, dependent contrasts. Nonlinear AR favors KAHM over both geometric alternatives under every head and seed; its block-direct MSE 0.00456 also improves on

ridge AR 0.00511 and quadratic regression 0.00523. On PenDigits, the reductions relative to RBF are 15.6%, 17.2%, and 17.9% across heads, with all five seeds favorable. However, Frobenius-direct KAHM MSE 0.00504 exceeds ridge AR 0.00184 and quadratic regression 0.00102. Affine geometry is stronger under Frobenius and rollout heads in every seed on both oscillators. Favorable mean contrasts can also oppose most seed-specific signs.

For each synthetic task and seed, a joint 0.05 allocation covers all preprocessing arms and six planned paired geometry comparisons. Only one of the 180 comparisons certifies positive improvement: Van der Pol seed 5703, KAHM rollout versus RBF rollout, with lower MSE-improvement endpoint 0.00107. The remaining comparisons are inconclusive. PenDigits comparative effects remain descriptive because the same held-out cohort was examined earlier. The supplement retains the complete fixed-dimension interpretation and confidence accounting.

11.4 Preprocessing is part of the representation

Training-standardized histories are the original default. The primary alternative fits coordinate minima a_j and maxima b_j on training data and freezes the map

$$z_j(x) = \begin{cases} 2(x_j - a_j)/(b_j - a_j) - 1, & b_j > a_j, \\ 0, & b_j = a_j. \end{cases} \tag{34}$$

Future extrema may fall outside $[-1, 1]$; clipping is a distinct representation. Targets and their units are unchanged. Section 5 explains why the full reconstruction-and-affinity map is not invariant to this scaling.

The original-seed follow-up is descriptive; a frozen confirmation protocol uses three new streams on each of six synthetic tasks. Duffing MSE decreases by 7.07% and 7.29% in the original and confirmation stages, whereas logistic MSE increases by 2.65% and 6.97% and Van der Pol MSE increases by 3.82% and 2.26%. The nonlinear-AR effect changes sign. On the fixed PenDigits cohort, soft-KAHM MSE decreases by 19.79% and 19.65%, but hard assignments worsen by 3.96% and 4.14%. Conventional regressors remain stronger. These outcomes support selecting preprocessing on development data, rather than prescribing one transformation for every task. The matched study above makes that selection before fresh calibration.

11.5 Natural-data calibration and its limits

PenDigits uses original unnormalized trajectories with a fixed acquisition-box transformation. Its 1751 eligible held-out digits belong to 13 inferred writer identifiers. Losses are averaged within writers. Even under an iid eligible-writer assumption, generic upper-RMSE endpoints near 1.56 certify none of the declared tolerances $0.2, 0.35, 0.5$. Completed-stroke eligibility defines a retrospective population, and repeated fitting seeds do not supply additional independent writers. The accuracy comparisons therefore remain descriptive.

The official Mini Speech Commands subset supplies 7178 eligible one-second clips in 1630 documented recording identifiers (Warden, 2018; TensorFlow, 2026). A fixed split assigns 244/163/978/245 groups to training/validation/calibration/test. The causal observable is frame energy in decibels relative to full scale, floored at -60 dB and mapped to $[0, 1]$; multiplying normalized RMSE by 60 returns dB error for that floored observable. Histories contain 4 or 12 frames and horizons are 10, 40, and 120 ms. Losses weight identifiers equally after averaging their origins and

clips. The confidence statement assumes iid group-record bundles; identifiers and disjoint splits do not establish that population assumption.

Three fitting seeds compare 486 candidates per fit across eight families; all 24 validation-selected models are frozen before held-out evaluation. A joint 0.05 allocation covers the 72 selected-model/horizon upper bounds. All six learned families pass the all-horizon 18 dB criterion in every fit; persistence and the constant model do not. No family passes the 6 or 12 dB criterion. At 120 ms, mean test RMSE is 12.73 dB for KAHM, 12.51 for ridge AR, 12.47 for quadratic regression, and 12.39 for Gaussian-kernel regression, versus 14.70 for persistence. The conventional regressors also outperform KAHM at shorter horizons. The tolerances are declared resolution levels, not established audio-application requirements.

A later scaling analysis reuses this cohort and makes no new confidence claim. Min–max scaling lowers KAHM all-horizon MSE by 1.41%, but raw energy histories perform better still; the conventional regressors remain stronger than every KAHM scaling arm. Thus speech demonstrates informative upper calibration under its sampling model, while natural-data class exclusion and a general KAHM forecasting advantage remain unestablished. The complete observable, model grid, scoring-order record, and scaling results accompany the supplement and Appendix C.1.

12 Discussion and limitations

What is certified about a representation. Lower and upper risk bounds can establish both that a fitted forecast meets a tolerance and that every cheaper declared class fails it. This identifies minimum feasible representation cost even when upper-only selection returns the same predictor. The conclusion concerns a representation, metric, sensitivity budget, and evaluation law jointly. As (1) shows, it does not identify information loss separately from irreducible noise or sensitivity restrictions. The learned-KAHM study demonstrates these decisions on a recoverable signal and an independent future control; their prevalence on naturally acquired data remains open. Costs count declared features, not globally optimal forecasting complexity or runtime.

Population slack and calibration precision. Pairwise slack reflects conditional residual means and loose geometric inequalities (Proposition 2). Our consistency result assumes unconditional iid draws, bounded scalar targets, and compact representation support; conditional blocks can approach a smaller average of conditional optima. Larger blocks leave fewer independent scores, so finite-sample endpoints need not improve.

Fresh calibration conditional on 40 frozen KAHM banks connects the hierarchy to learned geometry (Section 10). In the independent-future control, $b = 8, 16$ exclude both classes at $n = 1024$ for every control bank while pairs do not. All lower methods assess the same scalar Lipschitz class. At that size stratification gives larger mean endpoints; full-sample empirical minimization resolves every control at $n = 256$, its only tested sample size. Larger blocks add no cost decision in the repairable regime. The learned-hierarchy study uses KL lower means, whereas the learned-pair study uses empirical Bernstein; cross-study differences do not isolate block size. These are conditional evaluations of fixed banks, not new training repetitions or natural-data evidence. Stratification can also remove useful contrasts, and a sharp Lipschitz bound can remain loose for the affine readout. Zero or unresolved lower bounds establish neither class adequacy nor predictive sufficiency.

What refinement guarantees. Retaining old blocks preserves earlier feasible forecasts, and exact duplication leaves the block-norm class unchanged. A ridge-regularized fit may still change. Neither

nesting nor a reduced obstruction guarantees improvement of a fitted predictor. Residual correlation gives a sufficient population condition when budget remains, while the distance-expansion result is necessary. Neither guarantees that unsupervised KAHM clustering discovers a sufficient state. The experiments freeze candidates before calibration; sequential refinement with fresh evaluation blocks remains untested.

Sampling populations and adaptation. Synthetic episodes are independently generated. PenDigits supplies only 13 eligible held-out writer identifiers, inferred from source fields; its generic endpoints are uninformative. Speech supplies documented recording identifiers and 978 calibration groups, permitting informative upper bounds under the explicit iid group-record assumption. Source identifiers and disjoint splits do not establish random population sampling. Eligibility rules based on completed recordings and source quality screening further restrict both real-data populations. Repeated fitting seeds on the same cohort measure fitting variation, not independent population replications.

Target ranges follow from the synthetic laws, the PenDigits acquisition box, or the fixed speech-energy transformation. Clipping predictions cannot bound loss for an unbounded target. The certificates do not transfer across changes in dynamics, sensors, noise, or origin laws. Original-seed scaling follow-ups reuse previously examined calibration, and their endpoints are nominal. Fresh synthetic studies use the stated independent streams; PenDigits and speech scaling follow-ups on examined cohorts remain descriptive. Confidence allocations cover each declared study, not the entire research campaign.

Accuracy and representation parameters. The matched comparisons support KAHM gains on nonlinear autoregression and against RBF geometry on the fixed PenDigits cohort. Affine-hull alternatives are stronger on the oscillators, while autoregressive, quadratic, and kernel regressors outperform KAHM on speech. Scaling to $[-1, 1]$ improves some tasks and worsens others under the fixed local cap of two. The dimension-policy study (Section 11.2) improves descriptive test error through validation selection of the cap, while native cap twenty has mixed effects. All selected KAHM lower endpoints remain zero and the declared tolerance decisions are unchanged. Increasing local rank is therefore neither a universal accuracy prescription nor evidence of improved class exclusion. Local dimension and preprocessing both belong to the representation specification; their joint interaction is not established by these separately defined studies. Preprocessing must be treated as a development-selected model choice, with training extrema frozen before calibration; clipping future inputs is a distinct representation. These experiments assess the stated certificate framework and forecasting implementations. They establish neither universal KAHM superiority nor an accuracy frontier against broadly tuned neural forecasters. Execution times depend on hardware and implementation, and feature count alone does not determine computational cost.

13 Conclusion

Assessing a forecasting representation requires distinguishing the performance of a fitted decoder from the risk of its entire declared class. The normalized block hierarchy connects a pairwise obstruction to optimal scalar Lipschitz risk under the stated conditions. Calibration, stratification, and retained feature blocks turn this lower information into accuracy and minimum-cost decisions. Learned KAHM dictionaries provide a concrete implementation; the guarantees are general. Their

usefulness depends on population slack, independent sample size, and computational cost, while the experiments retain task-dependent accuracy and unresolved decisions as part of the evidence.

Code, data, and research transparency

The archived research repository at revision `d6d74d1` contains the reference forecasting, input-scaling, matched-geometry, lower-bound comparison, block-hierarchy, learned-KAHM decision, learned-geometry hierarchy, controlled-decision, speech, and speech-scaling studies in `extensions/certified_forecasting`. Each study has its own protocol, source hashes, result records, and reproduction commands. The separate dimension-policy study of Section 11.2 is supplied in the accompanying research extension `extensions/dimension_certificates`; it is not part of that historical revision. Its frozen schedule, source hashes, development/calibration/test seals, complete analysis tables, and independent audit identify the evaluated family and the order of data access. The accompanying numerical-records archive provides the production records and checksums. The manuscript source package includes the verified small analysis inputs and a generator for all dimension-study tables, so compilation does not require that numerical archive. Appendix C specifies the forecasting configurations and independent sampling units. The separately compiled supplement retains complete secondary experimental tables and the local KAHM implementation details; every proof of the main results remains in this manuscript. The reference studies' reproduction guide is `extensions/certified_forecasting/docs/REPRODUCIBILITY.md`. Public data are identified by source and checksum; download scripts preserve attribution and license information. Appendix C.1 describes the speech study's evaluation order and its implications for interpretation.

Use of AI tools. AI tools assisted with derivation exploration, implementation, literature retrieval, and manuscript preparation.

Appendix A. Convex fitting and a numerical optimality certificate

This appendix derives the projection used to fit the decoder and a computable upper bound on its training-objective suboptimality. That optimization bound is distinct from the population prediction-risk certificate: one measures how accurately a finite training problem was solved, and the other evaluates forecasting error on new episodes.

Let N be the number of training windows and $p_s = \sum_{b=1}^{s} C_b$ the total number of retained feature coordinates. Write $F = [F_1, \dots, F_s] \in \mathbb{R}^{N \times p_s}$ for the fixed feature matrix, where $F_b \in \mathbb{R}^{N \times C_b}$ contains block b. Let $\mathsf{Y} \in \mathbb{R}^{N \times q}$ contain the corresponding stacked targets, with $q = d_y H$ for d_y output coordinates at each of H horizons. Training windows may be dependent; optimization of this finite objective requires no independence assumption on its rows.

With intercept $a \in \mathbb{R}^q$ and coefficient matrix $W \in \mathbb{R}^{p_s \times q}$, solve

$$\min_{W,a} \frac{1}{N}\|\mathsf{Y} - \mathbf{1}_N a^\top - FW\|_{\mathrm{F}}^2 + \lambda\|W\|_{\mathrm{F}}^2 \quad \text{subject to} \quad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \le \kappa. \tag{35}$$

Here $\mathbf{1}_N \in \mathbb{R}^N$ is the all-ones column vector, $W_b \in \mathbb{R}^{C_b \times q}$ contains the rows associated with F_b, $\lambda \ge 0$ is the ridge coefficient, and $\kappa \ge 0$ is the block-norm budget. The intercept is neither penalized nor constrained. The loss sums over output coordinates and horizons, then averages over training

windows; there is no additional division by H or q. Clipping is applied after fitting. Thus (35) minimizes a convex objective before clipping, not the evaluated clipped empirical loss.

Define the row means $\bar{F} = N^{-1}\mathbf{1}_N^\top F$ and $\bar{\mathsf{Y}} = N^{-1}\mathbf{1}_N^\top \mathsf{Y}$, and the centered matrices

$$F_c = F - \mathbf{1}_N \bar{F}, \qquad \mathsf{Y}_c = \mathsf{Y} - \mathbf{1}_N \bar{\mathsf{Y}}.$$

For fixed W, minimizing over the intercept gives $a^\top = \bar{\mathsf{Y}} - \bar{F}W$. Substituting this expression eliminates the intercept. With I_{p_s} the $p_s \times p_s$ identity matrix, put

$$M = F_c^\top F_c/N + \lambda I_{p_s} \in \mathbb{R}^{p_s \times p_s}, \qquad C = F_c^\top \mathsf{Y}_c/N \in \mathbb{R}^{p_s \times q}, \qquad c = \|\mathsf{Y}_c\|_{\mathrm{F}}^2/N.$$

The reduced objective and its gradient are

$$J(W) = \operatorname{tr}(W^\top M W) - 2\langle W, C\rangle_{\mathrm{F}} + c, \qquad G(W) = \nabla J(W) = 2(MW - C).$$

The Frobenius inner product is $\langle A, B\rangle_{\mathrm{F}} = \operatorname{tr}(A^\top B)$. Since M is positive semidefinite, J is convex. Moreover,

$$\|G(W) - G(V)\|_{\mathrm{F}} \le 2\|M\|_{\mathrm{op}}\|W - V\|_{\mathrm{F}},$$

where $\|M\|_{\mathrm{op}} = \sup_{\|v\|_2=1} \|Mv\|_2$ is the operator norm. The implementation uses projected gradient steps with acceleration and monotone restarts. The following formulas assess feasibility and optimization accuracy without requiring a convergence-rate claim for that implementation.

Proposition 15 (Projection onto the block norm ball) *Let $\mathcal{C}_\kappa = \{W : \sum_b \|W_b\|_{\mathrm{F}} \le \kappa\}$ be the feasible coefficient set, and let $A \in \mathbb{R}^{p_s \times q}$ have the same row-block partition as W. Define $a_b = \|A_b\|_{\mathrm{F}}$ and let $\mathcal{P}_\kappa(A)$ denote the nearest matrix in $\mathcal{C}_\kappa$ in Frobenius norm. If $\sum_b a_b \le \kappa$, then $\mathcal{P}_\kappa(A) = A$. Otherwise, for $\kappa > 0$, choose the unique threshold $\theta \in (0, \max_b a_b)$ satisfying $\sum_b (a_b - \theta)_+ = \kappa$. For $\kappa = 0$, take $\theta = \max_b a_b$. The projection is*

$$\big[\mathcal{P}_\kappa(A)\big]_b = \begin{cases} (1 - \theta/a_b)_+ A_b, & a_b > 0, \\ 0, & a_b = 0. \end{cases} \tag{36}$$

In particular, $\mathcal{P}_0(A) = 0$. For every feasible W, the objective gap satisfies

$$0 \le J(W) - J^* \le \mathcal{D}(W) := \langle G(W), W\rangle_{\mathrm{F}} + \kappa \max_b \|G_b(W)\|_{\mathrm{F}}, \tag{37}$$

where $J^* = \min_{V \in \mathcal{C}_\kappa} J(V)$ is also the optimum value of (35)*, and $G_b(W)$ is the row block of the gradient corresponding to W_b.*

Proof For a block B_b constrained to have Frobenius norm $v_b \ge 0$, Cauchy–Schwarz gives

$$\|B_b - A_b\|_{\mathrm{F}}^2 = v_b^2 + a_b^2 - 2\langle B_b, A_b\rangle_{\mathrm{F}} \ge (v_b - a_b)^2.$$

When $a_b > 0$, equality is attained by taking $B_b = (v_b/a_b)A_b$. When $a_b = 0$, the squared distance is v_b^2 and its optimum has $v_b = 0$. Thus matrix projection reduces to the scalar problem

$$\min_{v_1,\dots,v_s \ge 0} \sum_{b=1}^{s} (v_b - a_b)^2 \quad \text{subject to} \quad \sum_{b=1}^{s} v_b \le \kappa.$$

If the original norms are feasible, the solution is $v_b = a_b$. If $\kappa = 0$, the only feasible vector is zero. In the remaining case the budget constraint is active. A multiplier 2θ for that constraint reduces each coordinate minimization to $\min_{v_b \geq 0}(v_b - a_b)^2 + 2\theta v_b$, whose solution is $v_b = (a_b - \theta)_+$. The sum of these coordinates is continuous and strictly decreasing from $\sum_b a_b$ to zero as θ increases from zero to $\max_b a_b$, so the required threshold exists and is unique. Rescaling each nonzero block gives (36).

The feasible set is nonempty and compact, so continuity of J ensures that J^* is attained. For the objective gap, convexity gives, for every feasible V,

$$J(W) - J(V) \leq \langle G(W), W - V \rangle_{\mathrm{F}}.$$

Also,

$$\langle G(W), V \rangle_{\mathrm{F}} \geq -\sum_b \|G_b(W)\|_{\mathrm{F}} \|V_b\|_{\mathrm{F}} \geq -\kappa \max_b \|G_b(W)\|_{\mathrm{F}}.$$

The last value is attainable by assigning the entire budget to a block with largest gradient norm, in the direction opposite to that gradient (or by $V = 0$ if the gradient is zero). Substitute an optimal V in the convexity inequality to obtain (37). ■

The implementation records coefficient norms, the gap $\mathcal{D}(W)$, iteration count, and whether the requested gap tolerance was reached. These are floating-point diagnostics, not interval-arithmetic guarantees. The held-out risk certificate requires the returned model to satisfy the Lipschitz budget used in calibration; it does not require an optimal training solution.

For a joint multihorizon fit, let $W_{b,h}$ be the columns of block W_b associated with the output at horizon h. Then

$$\sum_b \|W_{b,h}\|_{\mathrm{F}} \leq \sum_b \|W_b\|_{\mathrm{F}} \leq \kappa,$$

so the joint budget also bounds each horizon decoder's sensitivity. For scalar outputs, $W_{b,h}$ is the column corresponding to h in the increasingly ordered horizon list, and its Frobenius norm is its Euclidean norm. Although the block constraint is invariant to exact feature duplication, the ridge term $\lambda\|W\|_{\mathrm{F}}^2$ is not: splitting coefficients among duplicate blocks changes that penalty. The duplicate-block optimization test therefore uses $\lambda = 0$.

A.1 Matrix-free optimization of scalar block scores

For a block of b observations, orient each edge $e = (i, j)$ of the complete graph with $i < j$. Let B have columns $e_i - e_j$, where e_i is coordinate vector i, and set $c_e = \kappa d(U_i, U_j)$. The constraints are $|\mathsf{B}^\top v| \leq c$. Positive and negative parts of any signed edge vector α give nonnegative multipliers for the two constraint directions. Minimizing the resulting Lagrangian over v yields the feasible dual value

$$D(\alpha) = \alpha^\top \mathsf{B}^\top Y - \sum_e c_e |\alpha_e| - \tfrac{1}{4}\|\mathsf{B}\alpha\|_2^2 \leq \min_{|\mathsf{B}^\top v| \leq c} \|Y - v\|_2^2.$$

Thus no convergence assumption is required for a finite signed dual to supply a lower value. The implementation minimizes $-D$ by accelerated proximal gradient with componentwise soft thresholding (Beck and Teboulle, 2009). Since $\mathsf{B}\mathsf{B}^\top = bI - \mathbf{1}\mathbf{1}^\top$, the smooth negative-dual term has gradient Lipschitz constant $b/2$, giving step size $2/b$. Incidence products use indexed sums, avoiding a dense constraint matrix.

To obtain a primal upper value from an approximate vector z, evaluate

$$v_i^- = \min_j \{z_j + \kappa d(U_i, U_j)\}, \qquad v_i^+ = \max_j \{z_j - \kappa d(U_i, U_j)\}.$$

The triangle inequality makes both envelopes κ-Lipschitz on the sampled support. Their average is feasible, as is a common translation by its mean residual. Clipping to the observed target interval preserves feasibility and cannot increase squared error. This construction remains valid for duplicate or nearly coincident feature rows. Primal values quantify the numerical gap; only guarded dual values enter lower calibration.

Each batched program follows its own stopping criterion with a fixed iteration limit; another block's convergence does not alter its iterates. The recorded dual and primal guards address floating-point roundoff numerically and are not interval-arithmetic verification. Early-stopped programs and their gaps are retained. The block score divides the lower squared-error sum by $b - 1$; the full-sample empirical-minimum comparator instead divides by its sample size, as required by Appendix B.4.

Appendix B. Comparison certificates and bounded-loss calibration

This appendix derives the comparison bounds used in the lower-certificate experiments. Each formula applies to independent, identically distributed calibration units and a model class fixed before calibration. The comparison examines the statistical information and computational cost of these established concentration methods relative to the pair witness.

B.1 Relation to an inverse-contrastive population bound

Let (U, Y) and (U', Y') be independent copies of a representation and its scalar target, with $\mathbb{E}Y^2 < \infty$. Use the same representation pseudometric d and sensitivity budget $\kappa > 0$ in both bounds below. Write W_κ for (2) with constant conditioning variable A. For a threshold $a > 0$, define $F = \{|Y - Y'| > a\}$, $\rho = \Pr(F)$, and $M = \mathbb{E}[\mathbf{1}_F d(U, U')^2] \le \varepsilon < \infty$. Put $v = (|Y - Y'| - \kappa d(U, U'))_+$. Since $\mathbf{1}_F|Y - Y'| \le \mathbf{1}_F v + \kappa \mathbf{1}_F d(U, U')$, the L^2 triangle inequality gives $\|\mathbf{1}_F v\|_{L^2} \ge (\|\mathbf{1}_F|Y - Y'|\|_{L^2} - \kappa\sqrt{M})_+$. On F, the target difference exceeds a, so the first norm is at least $a\sqrt{\rho}$. Therefore

$$\begin{aligned} W_\kappa &\ge \tfrac{1}{2}\|\mathbf{1}_F v\|_{L^2}^2 \ge \tfrac{1}{2}(a\sqrt{\rho} - \kappa\sqrt{M})_+^2 \\ &\ge \tfrac{1}{2}(a\sqrt{\rho} - \kappa\sqrt{\varepsilon})_+^2. \end{aligned} \tag{38}$$

The inverse-contrastive-loss hypothesis of Akash et al. (2021, Lemma 2 and its appendix proof) bounds M by ε. In their continuous-target setting and positive regime $\varepsilon < a^2\rho^2/\kappa^2$, their displayed population lower bound is $(a\rho - \kappa\sqrt{\varepsilon})^2/4$; it is no larger than (38) because $\sqrt{\rho} \ge \rho$. Thus the earlier paired-residual argument already implies a prediction-risk obstruction. The positive-part discrepancy gives the stronger population bound above for the same law. A comparison of finite-sample confidence bounds additionally requires the same sampling assumptions and error probability allocations.

B.2 A range-aware centered lower bound

For this scalar comparator, let $U, Y \in \mathbb{R}$ have known interval ranges of lengths D_U, D_Y, and use $d(u, u') = |u - u'|$. From $n \ge 2$ independent, identically distributed observations (U_i, Y_i), define

sample means $\bar{U} = n^{-1}\sum_i U_i$ and $\bar{Y} = n^{-1}\sum_i Y_i$, and sample standard deviations

$$s_U = \left(\frac{1}{n-1}\sum_{i=1}^{n}(U_i - \bar{U})^2\right)^{1/2}, \qquad s_Y = \left(\frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2\right)^{1/2}.$$

Also let $\widehat{v}_Y = n^{-1}\sum_i (Y_i - \bar{Y})^2$, which uses denominator n rather than $n-1$. Define $c_\alpha = \sqrt{2\log(1/\alpha)/(n-1)}$. For a total failure probability $\eta \in (0,1)$, use $\alpha = \eta/3$ if $D_U > 0$ and $\alpha = \eta/2$ if $D_U = 0$, and set

$$\sigma_Y^- = \max\left\{(s_Y - D_Y c_\alpha)_+,\ \sqrt{\left(\widehat{v}_Y - D_Y^2\sqrt{\frac{\log(1/\alpha)}{2n}}\right)_+}\right\},$$

$$\sigma_U^+ = \begin{cases}\min\{s_U + D_U c_\alpha, D_U/2\}, & D_U > 0,\\ 0, & D_U = 0,\end{cases} \qquad L_{\mathrm{ctr}} = (\sigma_Y^- - \kappa\sigma_U^+)_+^2. \tag{39}$$

Then $L_{\mathrm{ctr}} \le R_\kappa^*$ with probability at least $1-\eta$, where R_κ^* is the infimum risk over scalar κ-Lipschitz functions of U. Both target lower bounds hold on the same confidence event, which permits taking their maximum.

To verify this, rescale Maurer and Pontil (2009, Theorem 10) to obtain the one-sided events $\sqrt{\mathrm{Var}(Y)} \ge s_Y - D_Y c_\alpha$ and $\sqrt{\mathrm{Var}(U)} \le s_U + D_U c_\alpha$, each with probability at least $1-\alpha$. Also, $\widehat{v}_Y = \inf_a n^{-1}\sum_i (Y_i - a)^2$, where a may be restricted to the known target interval. Replacing one observation changes this infimum by at most D_Y^2/n, and $\mathbb{E}\widehat{v}_Y = (1-1/n)\,\mathrm{Var}(Y) \le \mathrm{Var}(Y)$. For $D_Y > 0$, the bounded-differences inequality consequently gives

$$\Pr\left\{\widehat{v}_Y - \mathbb{E}\widehat{v}_Y > D_Y^2\sqrt{\frac{\log(1/\alpha)}{2n}}\right\} \le \alpha.$$

This proves the second target lower bound; for $D_Y = 0$ it is identically zero. The deterministic interval bound $\mathrm{Var}(U) \le D_U^2/4$ follows by bounding the mean squared distance to the interval midpoint, and justifies the cap. A union bound combines the two target statements and, when $D_U > 0$, the representation statement. On their common event, (4) with constant A gives $R_\kappa^* \ge (\sqrt{\mathrm{Var}(Y)} - \kappa\sqrt{\mathrm{Var}(U)})_+^2 \ge L_{\mathrm{ctr}}$. When the representation is structurally constant, its variance is exactly zero and requires no confidence correction. A zero sample variance alone does not justify that substitution.

B.3 One-sided relative-entropy bounds

For probabilities $r, p \in [0,1]$, define binary relative entropy

$$\mathrm{kl}(r\|p) = r\log(r/p) + (1-r)\log\{(1-r)/(1-p)\}.$$

Terms with zero numerator are zero. At $p = 0$ or $p = 1$, the value is infinite unless $r = p$, in which case it is zero. For $c > 0$, let $\mathrm{kl}_-(r,c)$ and $\mathrm{kl}_+(r,c)$ be the smallest and largest $p \in [0,1]$ satisfying $\mathrm{kl}(r\|p) \le c$. These endpoints lie respectively below and above r. They are nondecreasing functions of r and can be found by scalar bisection. In particular, $\mathrm{kl}_-(0,c) = 0$ and $\mathrm{kl}_+(1,c) = 1$.

Let $V_1, \ldots, V_n$ be independent and identically distributed in $[0, B]$, with $B > 0$, mean μ, and sample mean $\bar{V}$. For $\eta \in (0, 1)$ and $c = \log(1/\eta)/n$, each of

$$B \operatorname{kl}_-(\bar{V}/B, c) \le \mu, \qquad \mu \le B \operatorname{kl}_+(\bar{V}/B, c) \tag{40}$$

has failure probability at most η. A joint two-sided claim allocates failure probability to both directions. If $B = 0$, both endpoints are zero. The scores need not be Bernoulli observations.

For completeness, convexity of the exponential gives, for every real t,

$$\mathbb{E}e^{tV_1/B} \le 1 - \mu/B + (\mu/B)e^t.$$

Write $p = \mu/B$. Independence and the exponential Markov inequality give, for $z \in (p, 1]$,

$$\Pr\{\bar{V}/B \ge z\} \le \inf_{t \ge 0} e^{-tnz}(1 - p + pe^t)^n = e^{-n \operatorname{kl}(z\|p)}.$$

For $z \in [0, p)$, the same calculation with $t \le 0$ bounds $\Pr\{\bar{V}/B \le z\}$ by the same expression (Hoeffding, 1963). If $p \in \{0, 1\}$, boundedness makes every score constant and the assertions hold directly. For $0 < p < 1$, the divergence increases as z moves away from p on either side. Inverting these two one-sided bounds at probability η proves (40). Applying the upper endpoint to losses of a frozen forecaster gives an alternative to (22), under the same sampling and range assumptions. An episode average is one bounded score. Choosing between different concentration methods after seeing calibration requires a joint allocation, just as in Section 3.

B.4 A certified empirical-minimum comparator

An empirical-minimum lower bound provides a comparison with structural certificates that avoid optimization on calibration observations (Kumar and Kargaran, 2026, Section 2.5). Let $\mathcal{G}$ be a fixed nonempty class with losses $\ell_g(X, Y) \in [0, B]$, $B > 0$, and define the following quantities from independent, identically distributed observations (X_i, Y_i) of the target law:

$$R^* = \inf_{g \in \mathcal{G}} \mathbb{E}\ell_g(X, Y), \qquad a_n = \inf_{g \in \mathcal{G}} \frac{1}{n} \sum_i \ell_g(X_i, Y_i).$$

Assume a_n is measurable. Let $\eta \in (0, 1)$ be the allocated failure probability. If a computed measurable value d_n satisfies $d_n \le a_n$ for every calibration sample, put $q_n = \max\{0, d_n/B\}$. Since $0 \le a_n \le B$, one has $0 \le q_n \le a_n/B \le 1$. Then

$$L_{\mathrm{ERM}} = B \operatorname{kl}_-\left(q_n, \frac{\log(1/\eta)}{n}\right) \le R^* \tag{41}$$

with probability at least $1 - \eta$.

Indeed, $a_n \le n^{-1} \sum_i \ell_g(X_i, Y_i)$ for every fixed g. For $t \ge 0$, the preceding moment-generating-function bound therefore gives

$$\mathbb{E}e^{tna_n/B} \le \inf_{g \in \mathcal{G}} \{1 - R(g)/B + (R(g)/B)e^t\}^n = \{1 - R^*/B + (R^*/B)e^t\}^n,$$

where $R(g) = \mathbb{E}\ell_g(X, Y)$. The equality follows because $1 - r + re^t$ is nondecreasing in r for $t \ge 0$ and the class has predictors with risks arbitrarily close to R^*. If $R^* = 0$, this moment-generating-function bound and $a_n \ge 0$ imply $a_n = 0$ almost surely. If $R^* = B$, the conclusion

follows from $L_{\mathrm{ERM}} \le B$. For $0 < R^* < B$, the exponential Markov argument yields $\Pr\{a_n/B \ge z\} \le \exp\{-n\,\mathrm{kl}(z\|R^*/B)\}$ for $z \in (R^*/B, 1]$. Its inversion bounds $B\,\mathrm{kl}_-(a_n/B, \log(1/\eta)/n)$ from above by R^* with probability at least $1-\eta$. Monotonicity in the first argument and $q_n \le a_n/B$ prove (41). Neither attainment of the population infimum nor a uniform complexity penalty is needed: the empirical infimum lies below the empirical risk of each fixed population competitor. The result lower-bounds the risk of every class member; it supplies neither an upper risk guarantee nor an excess-risk bound for a predictor fitted to these calibration observations.

For scalar Lipschitz regression on a finite calibration sample, the empirical infimum is a convex quadratic program (Gottlieb et al., 2017, Section III-A). Restricting predictions to the known target interval $[y_-, y_+]$ preserves the optimum, because clipping reduces loss and preserves the Lipschitz budget. Here $y_- \le y_+$ are known bounds with $Y \in [y_-, y_+]$ almost surely. Let $u_1, \dots, u_{m_U}$ be the $m_U \ge 1$ distinct observed representation values and let $v_j \in [y_-, y_+]$ be the fitted value at u_j. Set

$$w_j = \frac{\#\{i : U_i = u_j\}}{n}, \qquad t_j = \frac{1}{n}\sum_{i:U_i=u_j} Y_i, \qquad \nu = \frac{1}{n}\sum_i Y_i^2.$$

Here $\#$ counts sample indices, $j = 1, \dots, m_U$, and $U_i = \Gamma(X_i)$. Expanding the squared losses shows that the empirical risk is $\nu + \sum_j (w_j v_j^2 - 2t_j v_j)$. Minimize this expression subject to $v_j - v_k \le \kappa d(u_j, u_k)$ for every ordered pair (j,k) and $y_- \le v_j \le y_+$. These constraints are necessary for a global Lipschitz decoder. They are also sufficient: the function $g(u) = \min_j\{v_j + \kappa d(u, u_j)\}$ is measurable, κ-Lipschitz, and equals v_j at every observed node. Projecting g onto $[y_-, y_+]$ preserves those values and its Lipschitz constant. For ordered scalar inputs with absolute distance, constraints in both directions between adjacent nodes imply every pair constraint by summation.

Let $v = (v_1, \dots, v_{m_U})^\top$ and write the L_{con} pairwise and interval constraints as $Av \le b$, with $A \in \mathbb{R}^{L_{\mathrm{con}} \times m_U}$ and $b \in \mathbb{R}^{L_{\mathrm{con}}}$. For any nonnegative multiplier vector $\zeta \in [0,\infty)^{L_{\mathrm{con}}}$, a dual lower value is

$$d_{\mathrm{dual}}(\zeta) = \nu - \zeta^\top b - \sum_{j=1}^{m_U} \frac{((A^\top \zeta)_j - 2t_j)^2}{4w_j}. \tag{42}$$

Indeed, adding $\zeta^\top(Av - b)$ to the quadratic objective gives its Lagrangian. Completing the square in each v_j shows that the unconstrained minimum occurs at $v_j = (2t_j - (A^\top\zeta)_j)/(2w_j)$ and equals (42). Every observed node has positive frequency, so $w_j > 0$ and these divisions are defined. For feasible v, the added term is nonpositive. Hence weak duality gives $d_{\mathrm{dual}}(\zeta) \le a_n$, supplying the required d_n. A primal objective alone is an upper bound on a_n and cannot replace this dual value. The implementation records primal feasibility and the primal–dual gap; its floating-point checks do not constitute interval-arithmetic proofs.

Appendix C. Sampling laws and reproducibility

The complete episode equations, numerical integration scheme, comparison-model definitions, candidate grids, and data eligibility rules are retained in the supplement's experimental-protocol section. The repository provides the exact configuration and source snapshot for each study. Table 8 summarizes the units on which the main confidence statements depend.

The four reference synthetic generators use independent initialization and noise: harmonic and quasiperiodic observations are bounded trigonometric sums, nonlinear AR uses a hyperbolic-tangent recurrence, and logistic observations are $2u - 1$ for $u \in [0,1]$. The oscillator observable is

Study	Independent unit and target	Interpretation
Finite-support hierarchy	One iid scalar observation, $Y \in [-1, 1]$	Known population optima; the confidence event is per law/repetition.
Learned KAHM controls	One independent two-lag history/future episode, $Y \in [-0.85, 0.85]$	Constructed repairable and independent-future laws; all selected banks and heads are frozen.
Synthetic forecasting	One independently initialized length-72 trajectory, $Y \in [-1, 1]$	Origins may overlap; complete-episode losses are independent across trajectories.
Dimension-policy comparison	One length-72 trajectory for fitted loss; one disjoint episode pair for a lower witness	All 26 development repetitions and selected banks precede calibration; one event covers the study's 4,680 designated endpoints.
PenDigits	One eligible writer record; acquisition coordinates in $[0, 500]$ have a fixed transformation	Thirteen held-out writers; source identifiers do not establish iid population sampling.
Speech energy	One recording-group bundle; fixed floored energy in $[0, 1]$	978 calibration groups under the stated iid group-record assumption; repeated fitting seeds reuse groups.

Table 8: Sampling units and deterministic target ranges. Clipping a predictor alone does not bound an unbounded target. Every confidence statement is conditional on its declared fixed development choices and sampling law.

$0.95 \tanh(x) + \eta$, $|\eta| \leq 0.02$, under the specified Runge–Kutta episode law. These finite sampling procedures define the target risks; no stationarity or invariant-distribution claim is required. The archived half-step numerical check assesses integration consistency rather than proving a continuous-time error bound.

For every comparison, available histories and targets share the same declared origin law. Training-fitted transformations and geometry are frozen before calibration. The fixed-cap reference seeds use separate streams for training, validation, calibration, and testing. Later confirmation studies have their own numeric seed rules and source freezes. Per-study confidence allocations do not automatically yield one confidence event for the full campaign. The fixed-cap reference comparisons use local KAHM cap $r_{\max} = 2$. The dimension-policy comparison uses native construction (15), 4096 independent calibration episodes and 2048 disjoint episode-pair scores per repetition. The thirteen within-episode origins are averaged and do not increase these sample sizes. Supplement Section S9 specifies its model grid, whole-bank validation selection, and campaign-wide allocation; that allocation does not include the other studies in the paper.

C.1 Speech evaluation order and reused cohorts

The speech protocol freezes all 24 selected models before held-out scoring. The first execution scored calibration then test for each model, instead of completing calibration for every model first. A strict-order replay retained identical selected models and numerical outcomes; both execution records and their scope remain in the supplement and repository. The later speech-scaling comparison is descriptive on the same cohort and does not score calibration groups or assert a new risk certificate. PenDigits fitting-seed follow-ups likewise reuse the same held-out writers.

Exact commands, dependencies, immutable result locations, and audit scope are listed in `docs/REPRODUCIBILITY.md` within the extension. Reproduction runners write new directories rather than overwrite retained outcomes. The main and supplementary PDFs are built together by `scripts/build_manuscript.py`; it resolves cross-document references without changing the retained experimental sources. All theorem and proposition proofs remain in the main manuscript.

Appendix D. Exact-risk benchmark and centered comparator

This benchmark has an analytically known optimal risk and fitted-model risk, so certificate coverage can be checked without using a test estimate as population truth. Let $Z \in \{0, 1\}$ be a Bernoulli state with $\Pr(Z = 1) = p \in (0, 1)$, and let the scalar target be $Y = 2Z-1$. For a representation-separation parameter $a \in [0, 1]$, use

$$\Phi_a(0) = ((1 + a)/2, (1 - a)/2)^\top , \qquad \Phi_a(1) = ((1 - a)/2, (1 + a)/2)^\top .$$

Both vectors lie in the two-coordinate probability simplex. Their Euclidean distance is $\sqrt{2}a$, whereas the target separation is 2. For sensitivity budget $\kappa \geq 0$, define the unexplained separation $c = (2 - \kappa\sqrt{2}a)_+$. Proposition 3 then gives

$$R^* = p(1 - p)c^2.$$

Here R^* denotes the minimum MSE over the benchmark's constrained predictors. The two target values, representation family, and support are specified in advance by the benchmark design.

An independent training sample gives the empirical frequency $\widehat{p}$ of state 1. We choose the constrained empirical-risk minimizer whose predictions at the two states are

$$\widehat{f}(0) = -1 + \widehat{p}c, \qquad \widehat{f}(1) = 1 - (1 - \widehat{p})c. \tag{43}$$

These values lie in $[-1, 1]$ and have separation $2 - c = \min\{2, \kappa\sqrt{2}a\}$. The affine construction in Proposition 3 therefore realizes them within the budget; when $a = 0$, the two values coincide. If one state is absent from training, (43) selects one feasible empirical minimizer using the predeclared support. This convention specifies the prediction even at an unobserved state.

Conditional on the training sample, state 0 has prediction error $-\widehat{p}c$, and state 1 has error $(1-\widehat{p})c$. Consequently the exact fitted population risk is

$$R(\widehat{f}) = c^2\{(1 - p)\widehat{p}^2 + p(1 - \widehat{p})^2\} = R^* + c^2(\widehat{p} - p)^2. \tag{44}$$

For an independent pair of calibration states, the witness score equals $c^2/2$ when the states differ and zero otherwise. The mismatch probability is $2p(1 - p)$, so the expected score is exactly R^*. With n calibration observations, the pair lower bound uses $\lfloor n/2 \rfloor$ disjoint pairs. An odd final observation is excluded from pairing but remains in the fitted-risk and centered statistics. The reported calibration sizes are all even.

The centered comparator. For comparison, apply a centered certificate based on the target and representation variances, with Euclidean distance and a single reference class. Let $Z_1, \ldots, Z_n$ be independent calibration states, with $n \geq 4$, and $\bar{z} = n^{-1} \sum_{i=1}^n Z_i$ their empirical frequency. The output and representation variances, computed with denominator n, are

$$\widehat{s} = 4\bar{z}(1 - \bar{z}), \qquad \widehat{t} = 2a^2\bar{z}(1 - \bar{z}).$$

Their population counterparts are $\sigma^2 = 4p(1-p)$ and $\tau^2 = 2a^2p(1-p)$. The population centered bound is $(\sigma - \kappa\tau)_+^2 \le R^*$, using nonnegative square roots. In this two-state benchmark the inequality is an equality.

For a failure probability $\eta \in (0,1)$, define

$$\begin{aligned} r &= \log(2/\eta)/n, \\ V_\eta &= (\widehat{s} - 4\sqrt{r/2})_+, \\ T_\eta &= \min\{1, \widehat{t} + \sqrt{2r} + 1/n\}, \\ L_\eta^{\mathrm{ctr}} &= (\sqrt{V_\eta} - \kappa\sqrt{T_\eta})_+^2. \end{aligned} \tag{45}$$

Here V_η is a lower bound on σ^2, T_η is an upper bound on τ^2, and the superscript "ctr" identifies the centered risk bound. We show that $L_\eta^{\mathrm{ctr}} \le R^*$ with probability at least $1-\eta$.

First, $\mathbb{E}\widehat{s} = (1-1/n)\sigma^2 \le \sigma^2$, and changing one calibration observation changes $\widehat{s}$ by at most $4/n$. The bounded-differences inequality therefore gives

$$\widehat{s} - \mathbb{E}\widehat{s} \le 4\sqrt{r/2}$$

with probability at least $1 - \eta/2$. On this confidence event, $V_\eta \le \sigma^2$. Second, changing one observation changes $\widehat{t}$ by at most $2/n$, so the same inequality gives

$$\mathbb{E}\widehat{t} - \widehat{t} \le \sqrt{2r}$$

with failure probability at most $\eta/2$. Since $\mathbb{E}\widehat{t} = (1-1/n)\tau^2$ and $\tau^2 \le 1$,

$$\tau^2 \le \widehat{t} + \sqrt{2r} + \tau^2/n \le \widehat{t} + \sqrt{2r} + 1/n.$$

Capping the right side at one remains valid because $\tau^2 \le 1$. A union bound gives $V_\eta \le \sigma^2$ and $T_\eta \ge \tau^2$ jointly with probability at least $1-\eta$. The expression in (45) increases with its first variance argument and decreases with its second. Hence it is no larger than $(\sigma - \kappa\tau)_+^2 \le R^*$ on that event. The constant one is the general simplex-variance bound; it is conservative for this two-coordinate example.

In each Monte Carlo trial, the fitted-risk upper bound, pair-witness lower bound, and centered lower bound each receive failure probability $\delta/3$. For the centered bound, setting $\eta = \delta/3$ splits that allocation between its two internal variance bounds. The three reported directions therefore hold jointly with probability at least $1-\delta$ within a trial. No joint $1-\delta$ event is asserted across all trials, grid conditions, or separately reported experiments.

References

David Abel, David Hershkowitz, and Michael Littman. Near optimal behavior via approximate state abstraction. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 2915–2923, 2016. URL `https://proceedings.mlr.press/v48/abel16.html`.

Aditya Kumar Akash, Vishnu Suresh Lokhande, Sathya N. Ravi, and Vikas Singh. Learning invariant representations using inverse contrastive loss. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 6582–6591, 2021. doi: 10.1609/aaai.v35i8.16815. URL `https://ojs.aaai.org/index.php/AAAI/article/view/16815`.

Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *The Annals of Applied Statistics*, 19(2):1641–1662, 2025. doi: 10.1214/24-AOAS1998. URL https://arxiv.org/abs/2110.01052.

Francis R. Bach. Consistency of the group lasso and multiple kernel learning. *Journal of Machine Learning Research*, 9(40):1179–1225, 2008. URL https://jmlr.org/papers/v9/bach08b.html.

Amir Beck and Marc Teboulle. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM Journal on Imaging Sciences*, 2(1):183–202, 2009. doi: 10.1137/080716542. URL https://epubs.siam.org/doi/10.1137/080716542.

Avrim Blum and Anh Tuan Nguyen. Robustly reliable learning for regression. arXiv:2610.17381v2, 2026. URL https://arxiv.org/abs/2610.17381v2.

Peter Bühlmann and Bin Yu. Boosting with the L_2 loss: Regression and classification. *Journal of the American Statistical Association*, 98(462):324–339, 2003. doi: 10.1198/016214503000125. URL https://stat.ethz.ch/Manuscripts/buhlmann/boosting.rev5.pdf.

Stephan Clémençon, Igor Colin, and Aurélien Bellet. Scaling-up empirical risk minimization: Optimization of incomplete U-statistics. *Journal of Machine Learning Research*, 17(76):1–36, 2016. URL https://jmlr.org/papers/v17/15-012.html.

Gustav Conradie, Nicolas Boullé, Jean-Christophe Loiseau, Steven L. Brunton, and Matthew J. Colbrook. Trustworthy Koopman operator learning: Invariance diagnostics and error bounds. arXiv:2603.15091, 2026. URL https://arxiv.org/abs/2603.15091.

George Coote and Matthew J. Colbrook. Residual-guided dictionary learning for spectrally accurate Koopman approximation. arXiv:2606.29083, 2026. URL https://arxiv.org/abs/2606.29083.

Dafydd Evans and Antonia J. Jones. A proof of the Gamma test. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 458(2027):2759–2799, 2002. doi: 10.1098/rspa.2002.1010. URL https://doi.org/10.1098/rspa.2002.1010.

Carles Gelada, Saurabh Kumar, Jacob Buckman, Ofir Nachum, and Marc G. Bellemare. DeepMDP: Learning continuous latent space models for representation learning. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2170–2179, 2019. URL https://proceedings.mlr.press/v97/gelada19a.html.

Lee-Ad Gottlieb, Aryeh Kontorovich, and Robert Krauthgamer. Efficient regression in metric spaces via approximate lipschitz extension. *IEEE Transactions on Information Theory*, 63(8):4838–4849, 2017. doi: 10.1109/TIT.2017.2713820.

Vincent Guigues, Anatoli Juditsky, and Arkadi Nemirovski. Non-asymptotic confidence bounds for the optimal value of a stochastic program. *Optimization Methods and Software*, 32(5):1033–1058, 2017. doi: 10.1080/10556788.2017.1350177. URL https://arxiv.org/abs/1601.07592.

Masih Haseli and Jorge Cortés. Temporal forward–backward consistency, not residual error, measures the prediction accuracy of extended dynamic mode decomposition. *IEEE Control Systems Letters*, 7:649–654, 2023. doi: 10.1109/LCSYS.2022.3214476. URL https://par.nsf.gov/servlets/purl/10466905.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830. URL https://www.tandfonline.com/doi/abs/10.1080/01621459.1963.10500830.

Zeyuan Jin, Mohammad Khajenejad, and Sze Zheng Yong. Data-driven model invalidation for unknown Lipschitz continuous systems via abstraction. In *2020 American Control Conference*, pages 2975–2980, 2020. doi: 10.23919/ACC45564.2020.9147725. URL https://arxiv.org/abs/2003.11662.

Michael I. Jordan and Robert A. Jacobs. Hierarchical mixtures of experts and the EM algorithm. *Neural Computation*, 6(2):181–214, 1994. doi: 10.1162/neco.1994.6.2.181. URL https://doi.org/10.1162/neco.1994.6.2.181.

Vladimir Kostic, Pietro Novelli, Andreas Maurer, Carlo Ciliberto, Lorenzo Rosasco, and Massimiliano Pontil. Learning dynamical systems via Koopman operator regression in reproducing kernel hilbert spaces. In *Advances in Neural Information Processing Systems*, volume 35, pages 4017–4031, 2022. URL https://papers.nips.cc/paper_files/paper/2022/hash/196c4e02b7464c554f0f5646af5d502e-Abstract-Conference.html.

Mohit Kumar and Somayeh Kargaran. Prediction limits and Koopman closure of geometry-induced soft state abstractions. arXiv:2609.32652v3, 2026. URL https://arxiv.org/abs/2609.32652v3.

Mohit Kumar, Bernhard A. Moser, and Lukas Fischer. On mitigating the utility-loss in differentially private learning: A new perspective by a geometrically inspired kernel approach. *Journal of Artificial Intelligence Research*, 79:515–567, 2024. doi: 10.1613/jair.1.15071. URL https://jair.org/index.php/jair/article/view/15071.

Mohit Kumar, Alexander Valentinitsch, Magdalena Fuchs, Mathias Brucker, Juliana Bowles, Adnan Husakovic, Ali Abbas, and Bernhard A. Moser. Geometrically inspired kernel machines for collaborative learning beyond gradient descent. *Journal of Artificial Intelligence Research*, 83 (16), 2025. doi: 10.1613/jair.1.16821. URL https://jair.org/index.php/jair/article/view/16821.

Qianxiao Li, Felix Dietrich, Erik M. Bollt, and Ioannis G. Kevrekidis. Extended dynamic mode decomposition with dictionary learning: A data-driven adaptive spectral decomposition of the Koopman operator. *Chaos*, 27(10):103111, 2017. doi: 10.1063/1.4993854. URL https://arxiv.org/abs/1707.00225.

Jeff Linderoth, Alexander Shapiro, and Stephen Wright. The empirical behavior of sampling methods for stochastic programming. *Annals of Operations Research*, 142(1):215–241, 2006. doi: 10.1007/s10479-006-6169-8. URL https://jlinderoth.github.io/papers/Linderoth-Shapiro-Wright-06.pdf.

Wai-Kei Mak, David P. Morton, and R. Kevin Wood. Monte Carlo bounding techniques for determining solution quality in stochastic programs. *Operations Research Letters*, 24(1–2):47–56, 1999. doi: 10.1016/S0167-6377(98)00054-6. URL https://www.sciencedirect.com/science/article/pii/S0167637798000546.

Andreas Maurer and Massimiliano Pontil. Empirical Bernstein bounds and sample variance penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory*, 2009. URL https://arxiv.org/abs/0907.3740.

Daniel J. McDonald, Cosma Rohilla Shalizi, and Mark Schervish. Nonparametric risk bounds for time-series forecasting. *Journal of Machine Learning Research*, 18(32):1–40, 2017. URL https://www.jmlr.org/papers/v18/13-336.html.

Mario Milanese and Carlo Novara. Set membership identification of nonlinear systems. *Automatica*, 40(6):957–975, 2004. doi: 10.1016/j.automatica.2004.02.002. URL https://www.sciencedirect.com/science/article/pii/S0005109804000470.

Manuel M. Müller, Yuetian Luo, and Rina Foygel Barber. Are all models wrong? fundamental limits in distribution-free empirical model falsification. In *Proceedings of the 38th Conference on Learning Theory*, volume 291 of *Proceedings of Machine Learning Research*, pages 4271–4308, 2025. URL https://proceedings.mlr.press/v291/muller25a.html.

Roberto I. Oliveira, Paulo Orenstein, Thiago Ramos, and João Vitor Romano. Split conformal prediction and non-exchangeable data. *Journal of Machine Learning Research*, 25(225):1–38, 2024. URL https://www.jmlr.org/papers/v25/23-1553.html.

Satinder P. Singh, Tommi Jaakkola, and Michael I. Jordan. Reinforcement learning with soft state aggregation. In *Advances in Neural Information Processing Systems*, volume 7, pages 361–368, 1994. URL https://papers.neurips.cc/paper_files/paper/1994/hash/287e03db1d99e0ec2edb90d079e142f3-Abstract.html.

TensorFlow. Simple audio recognition: Recognizing keywords, 2026. URL https://www.tensorflow.org/tutorials/audio/simple_audio. Official Mini Speech Commands subset and tutorial. Accessed October 29, 2026.

Pete Warden. Speech commands: A dataset for limited-vocabulary speech recognition. *arXiv preprint arXiv:1804.03209*, 2018. URL https://arxiv.org/abs/1804.03209.

Matthew O. Williams, Ioannis G. Kevrekidis, and Clarence W. Rowley. A data-driven approximation of the Koopman operator: Extending dynamic mode decomposition. *Journal of Nonlinear Science*, 25(6):1307–1346, 2015. doi: 10.1007/s00332-015-9258-5. URL https://arxiv.org/abs/1408.4408.

Liming Wu and Sen Yang. Non-asymptotic estimates of the minimal risk in statistical learning. arXiv:2606.23295, 2026. URL https://arxiv.org/abs/2606.23295.